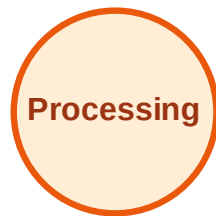


BFS Traversal

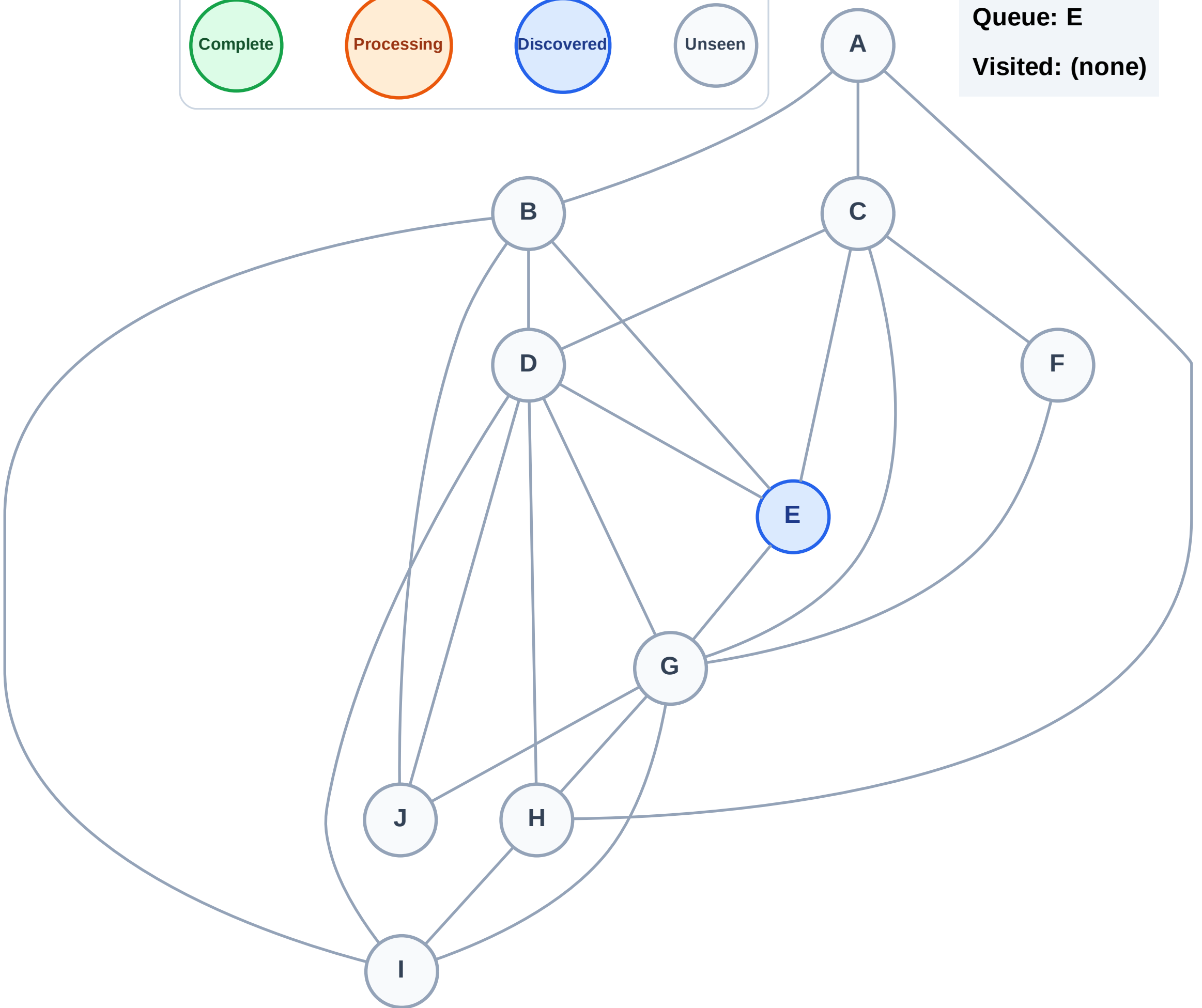
Step 1: Start at E

Legend



Queue: E

Visited: (none)



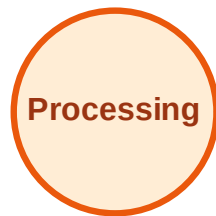
BFS Traversal

Step 2: Process node E

Legend



Complete



Processing



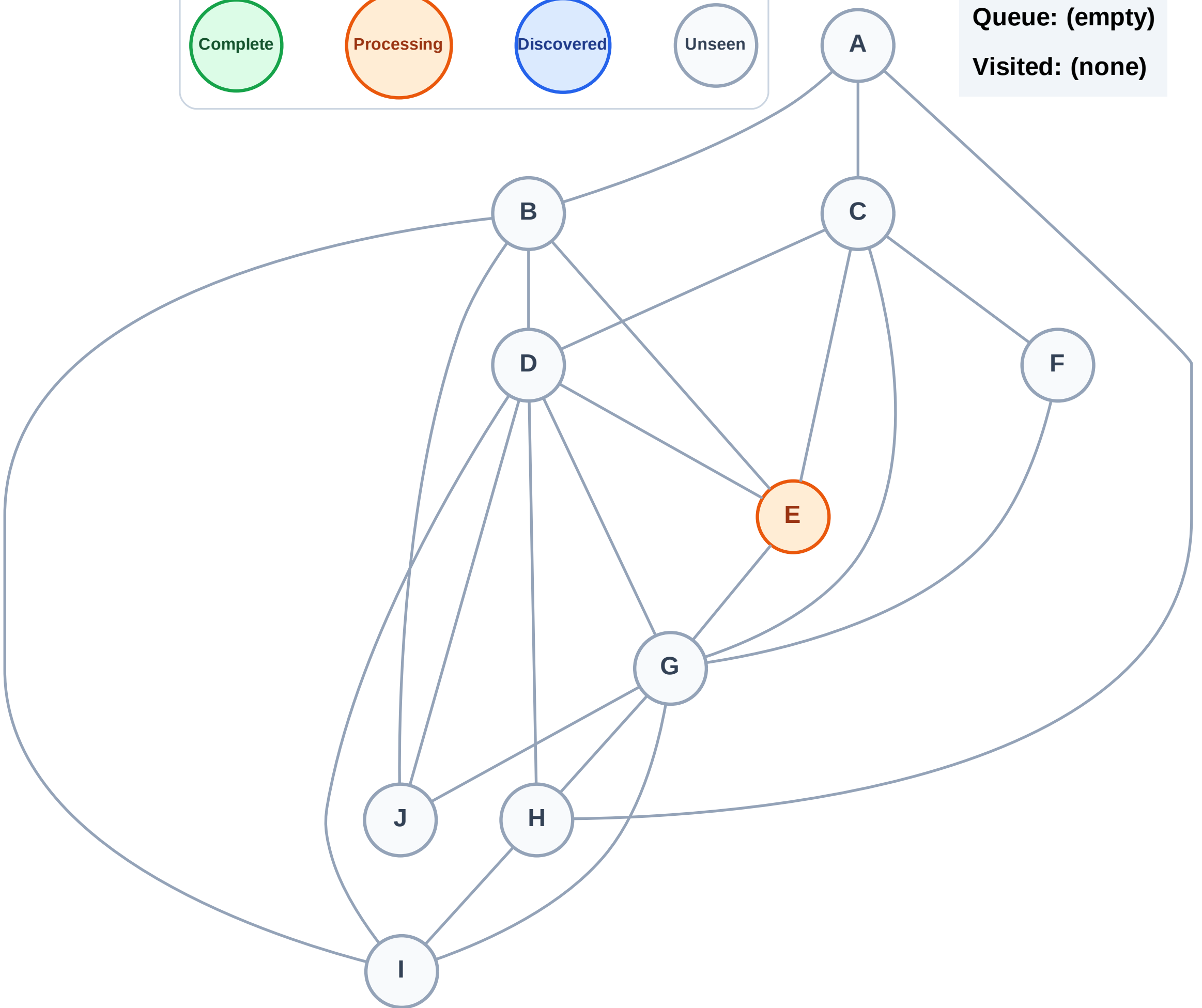
Discovered



Unseen

Queue: (empty)

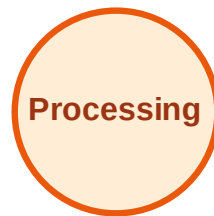
Visited: (none)



BFS Traversal

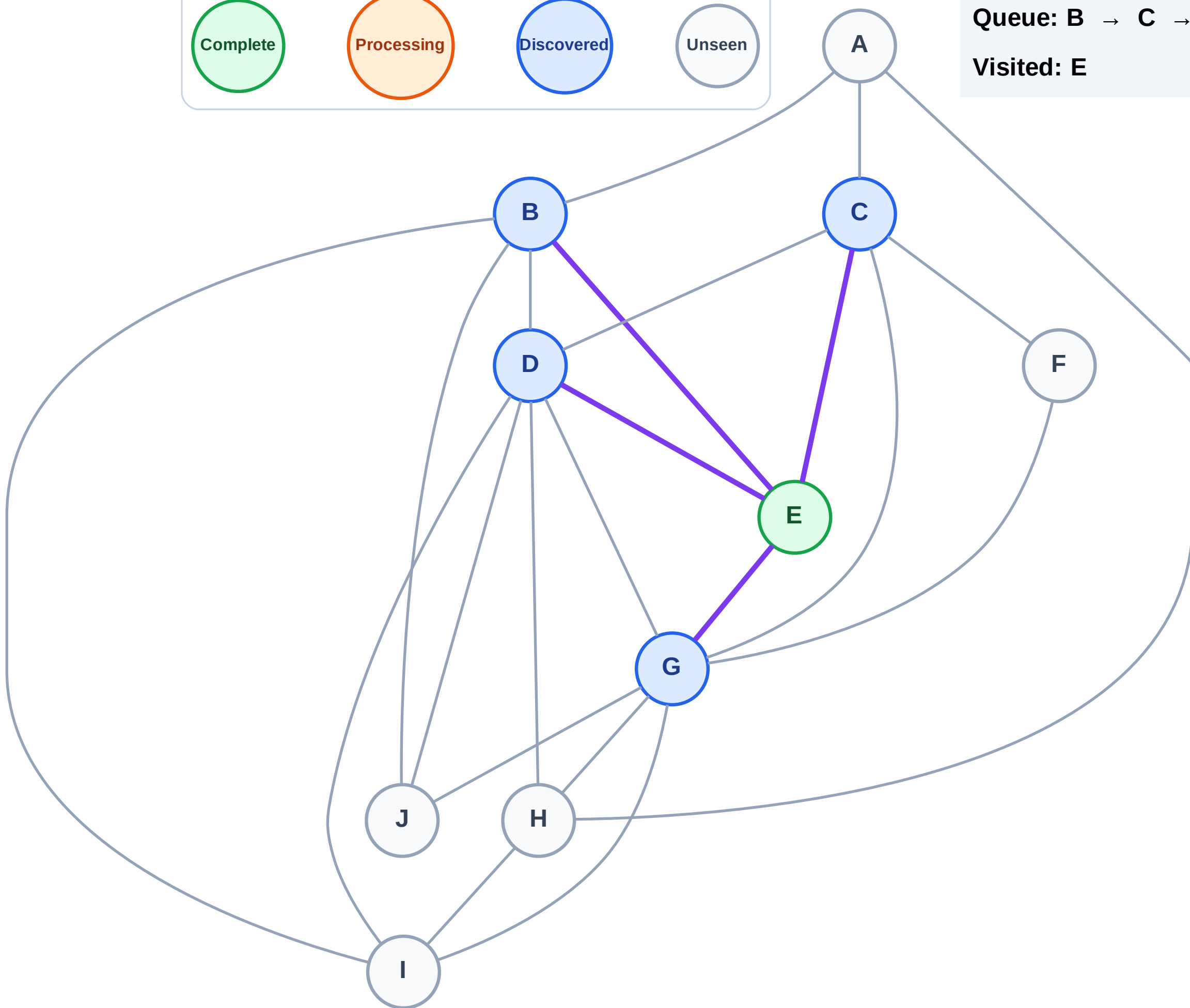
Step 3: Discover B, C, D, G from E

Legend



Queue: B → C → D → G

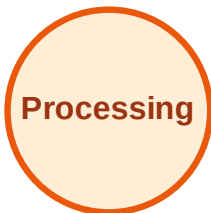
Visited: E



BFS Traversal

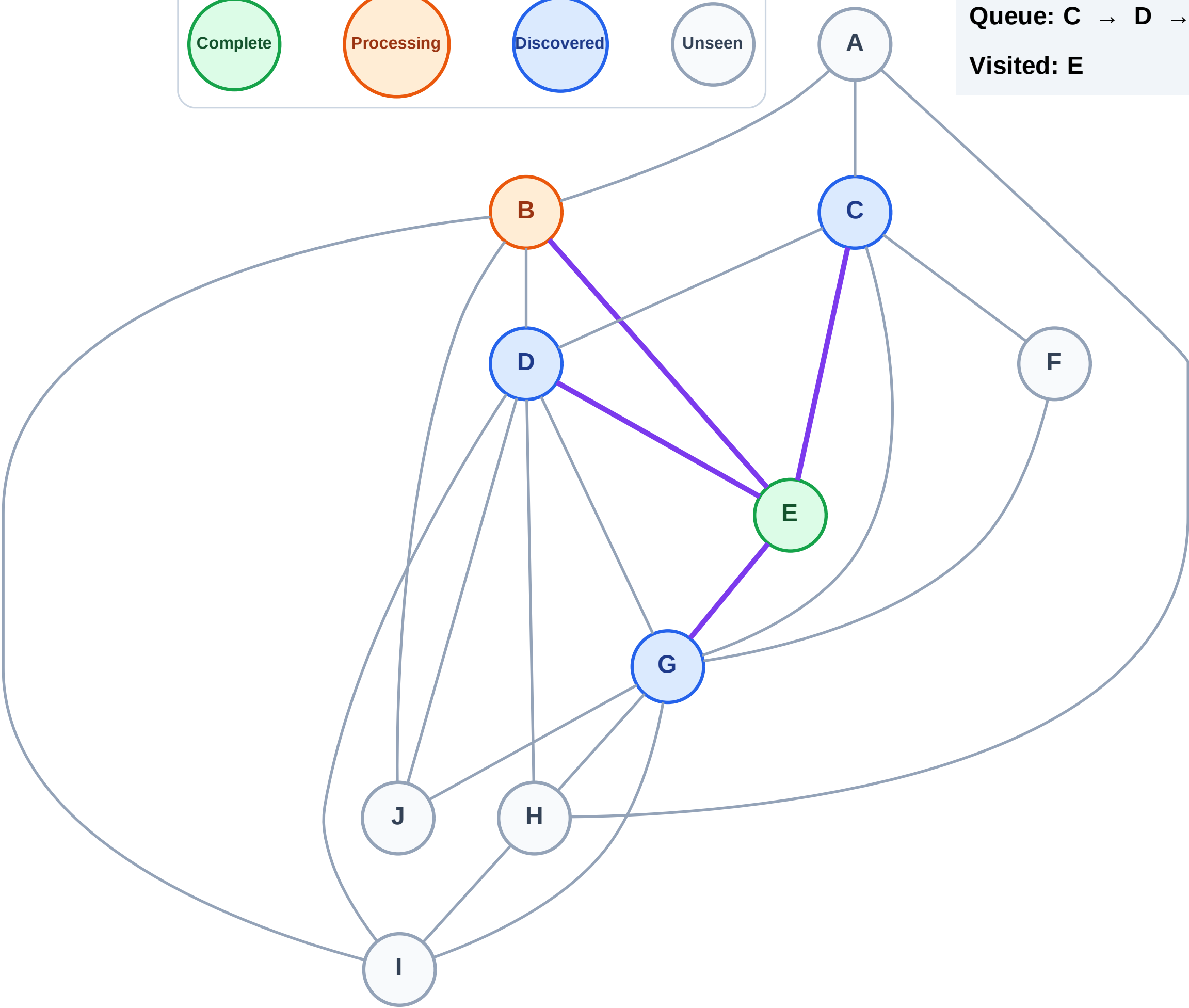
Step 4: Process node B

Legend



Queue: C → D → G

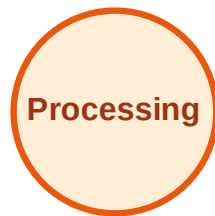
Visited: E



BFS Traversal

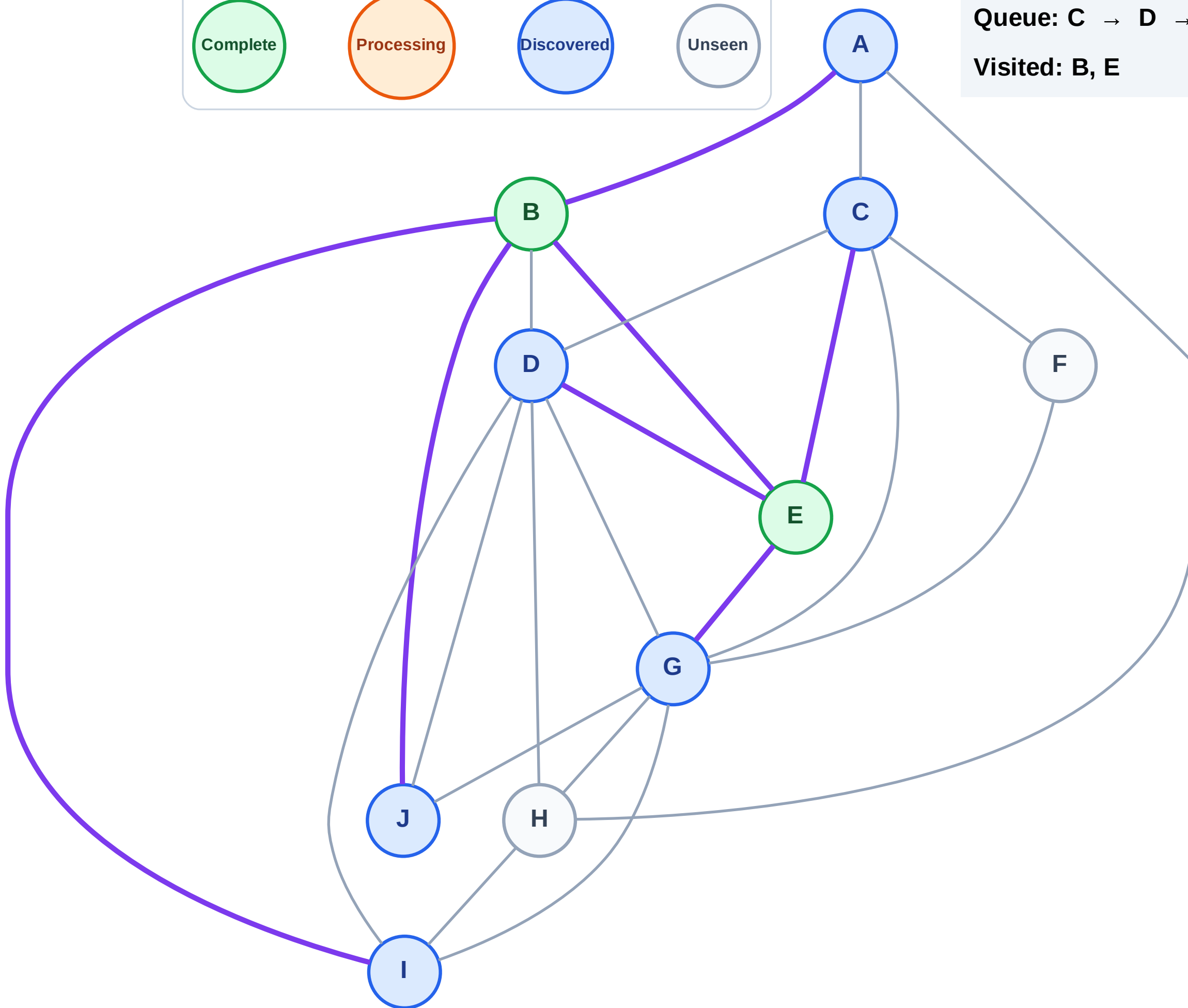
Step 5: Discover A, I, J from B

Legend



Queue: C → D → G → A → I → J

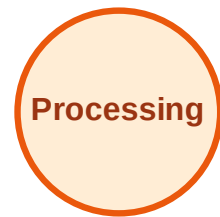
Visited: B, E



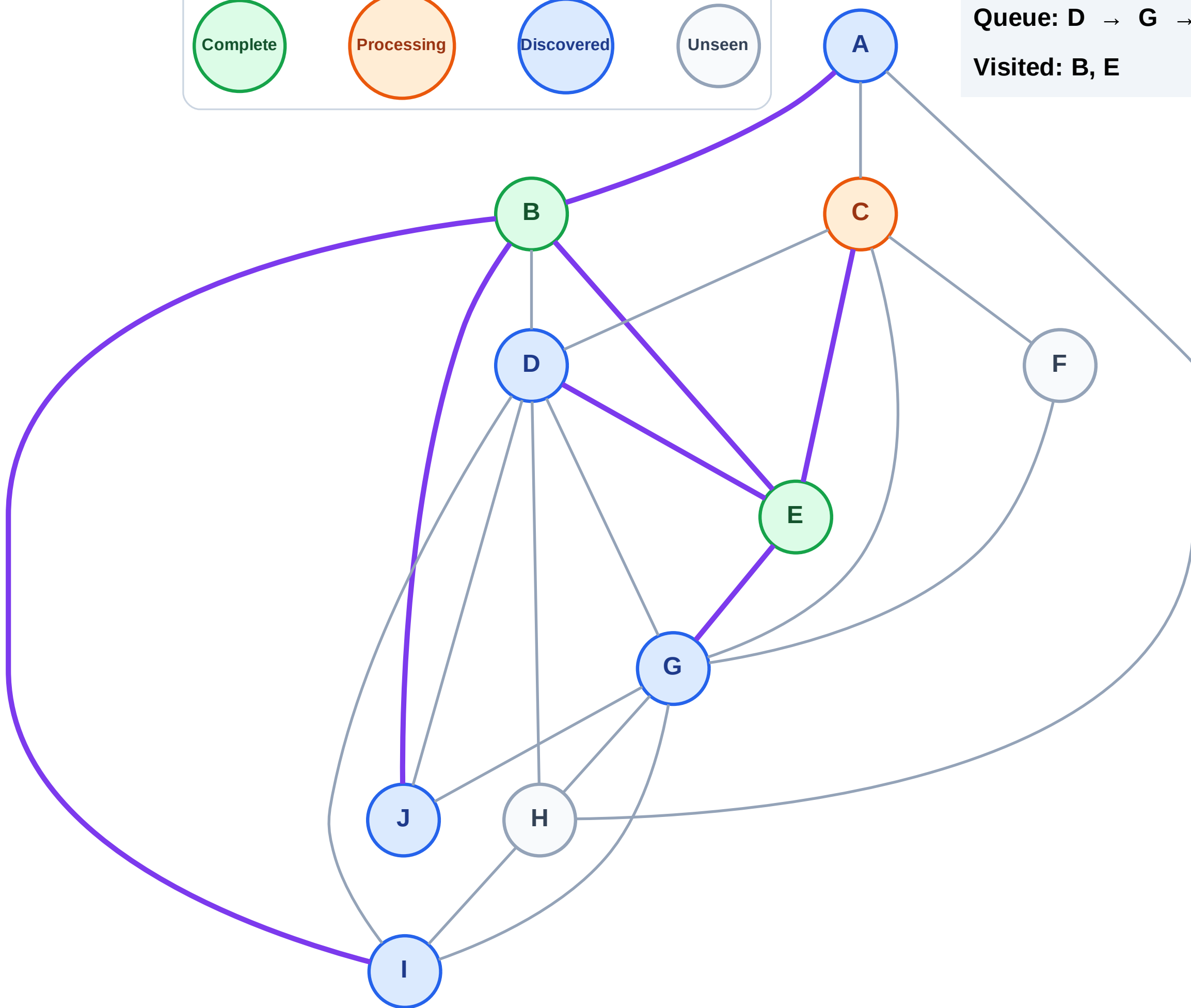
BFS Traversal

Step 6: Process node C

Legend



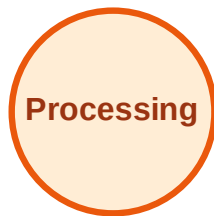
Queue: D → G → A → I → J
Visited: B, E



BFS Traversal

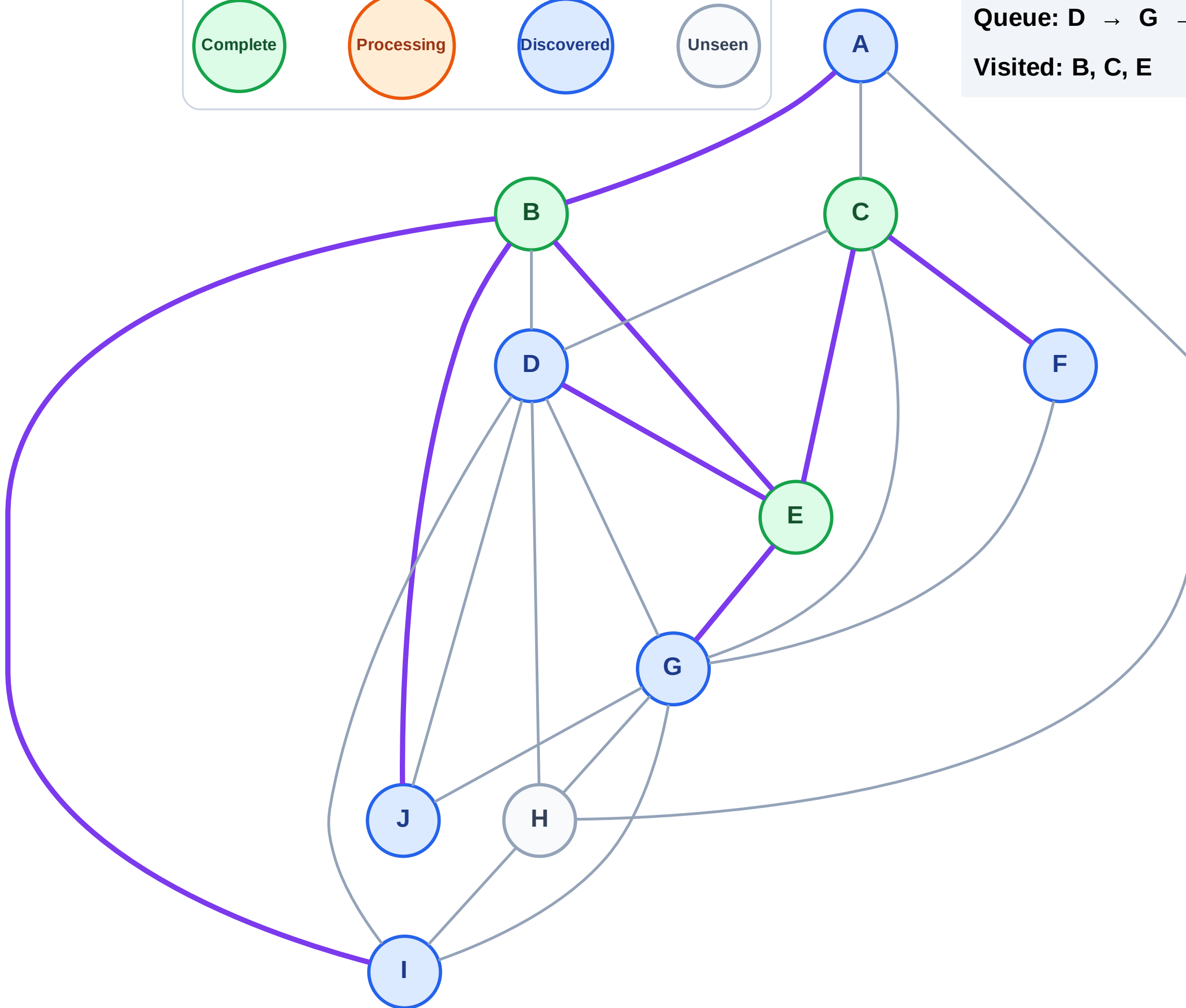
Step 7: Discover F from C

Legend



Queue: D → G → A → I → J → F

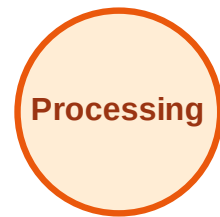
Visited: B, C, E



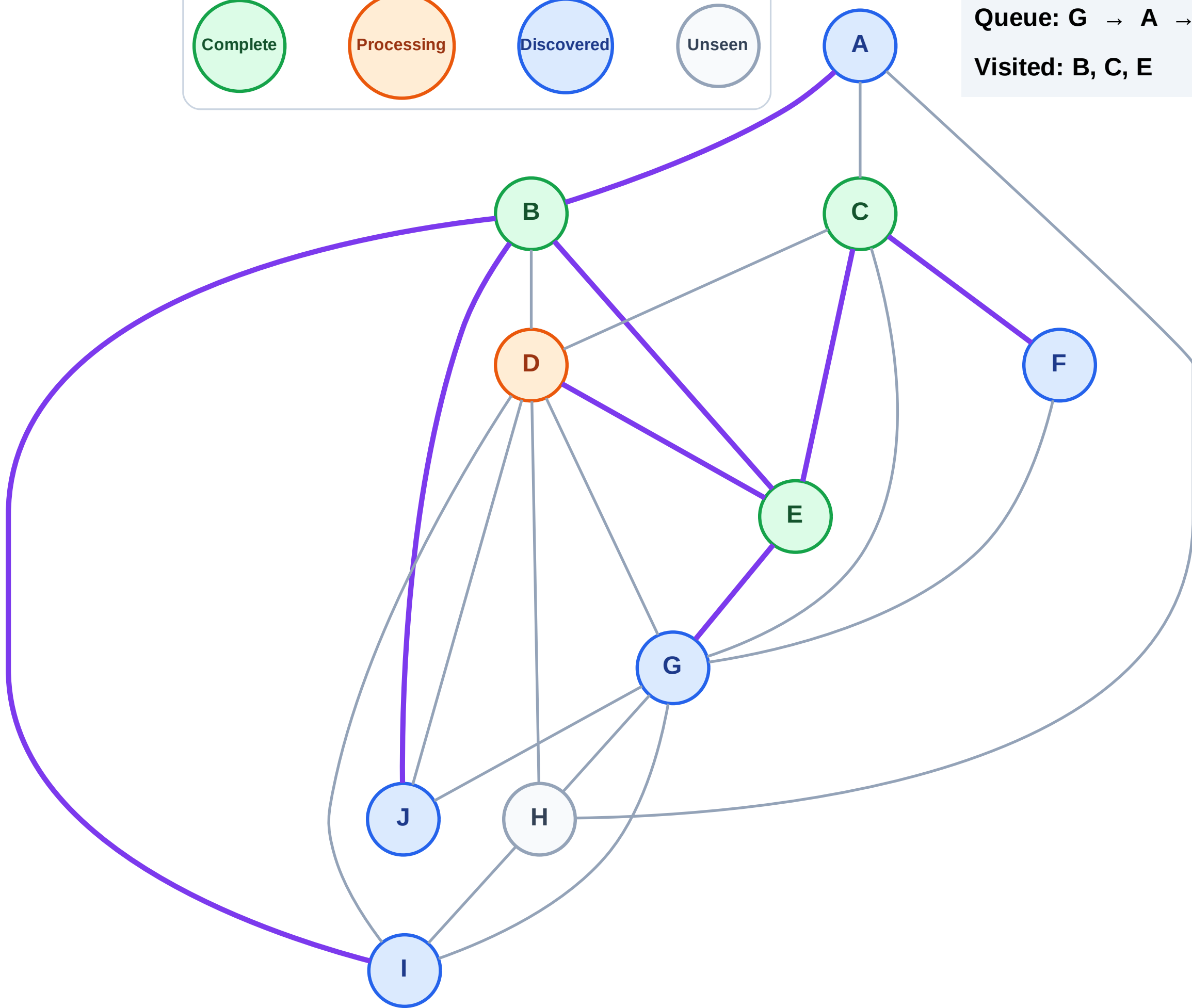
BFS Traversal

Step 8: Process node D

Legend



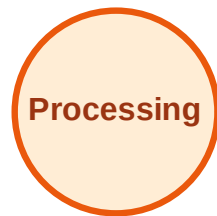
Queue: G → A → I → J → F
Visited: B, C, E



BFS Traversal

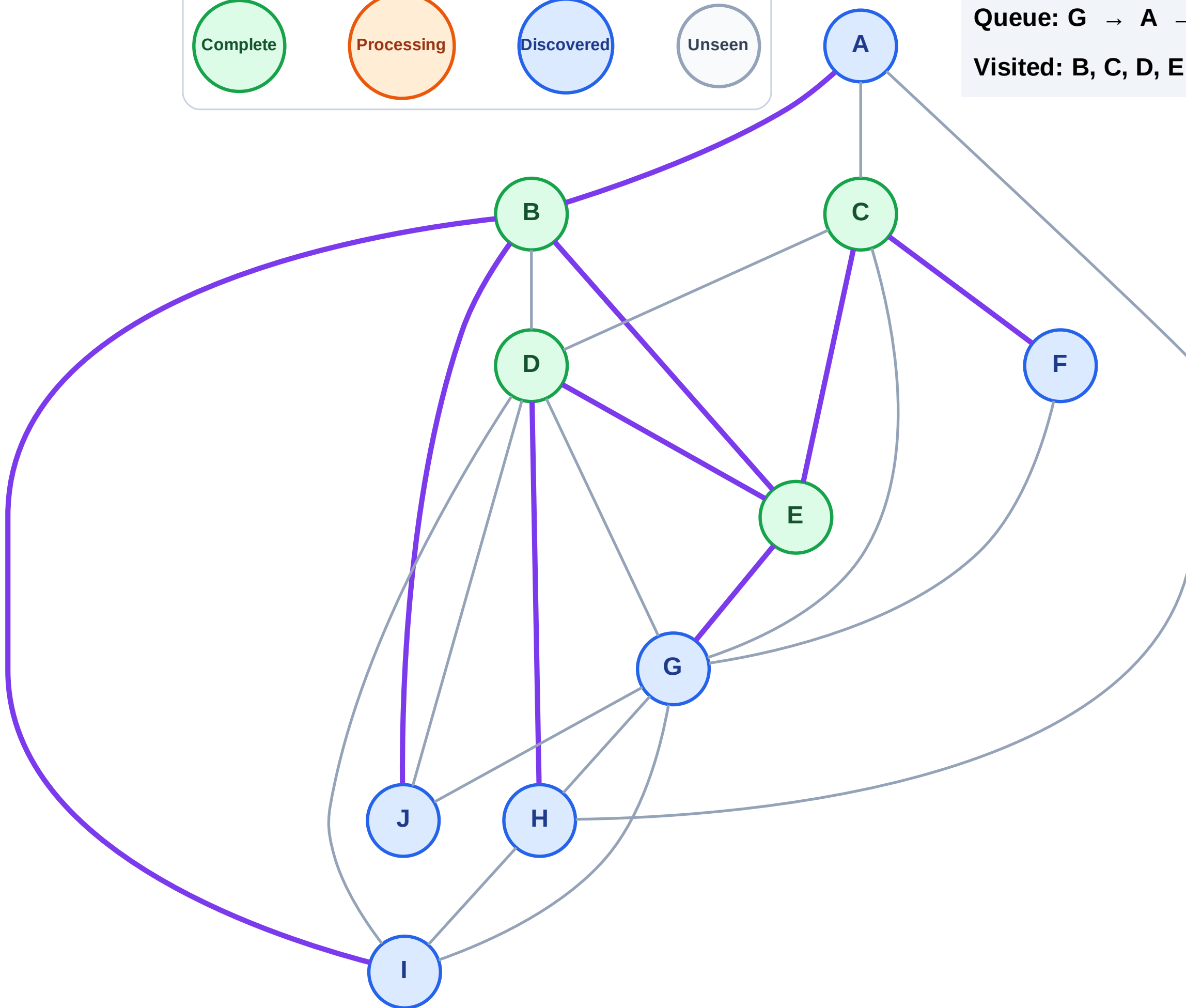
Step 9: Discover H from D

Legend



Queue: G → A → I → J → F → H

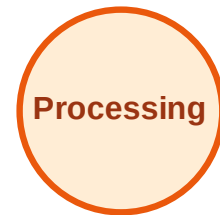
Visited: B, C, D, E



BFS Traversal

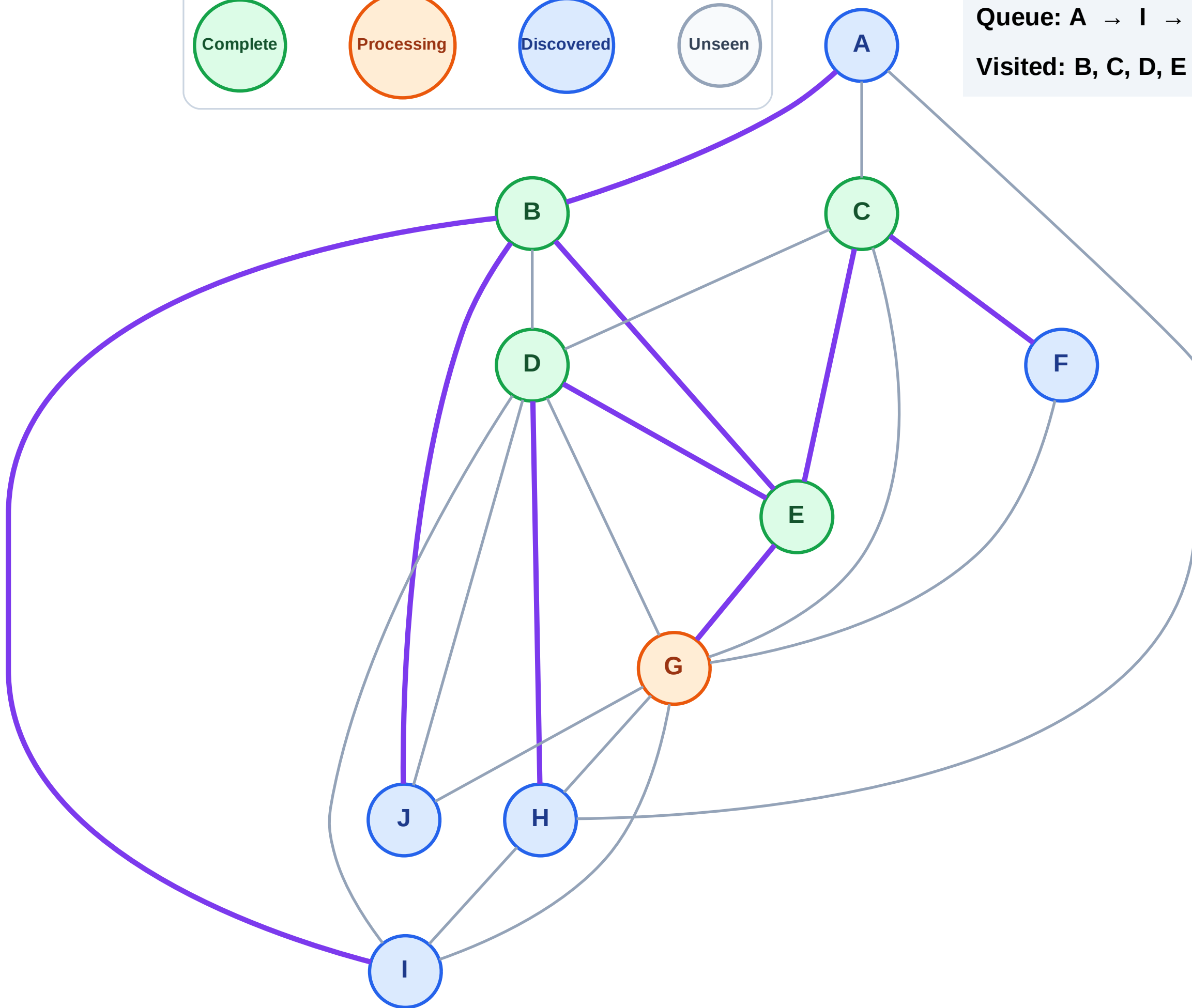
Step 10: Process node G

Legend



Queue: A → I → J → F → H

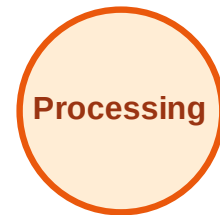
Visited: B, C, D, E



BFS Traversal

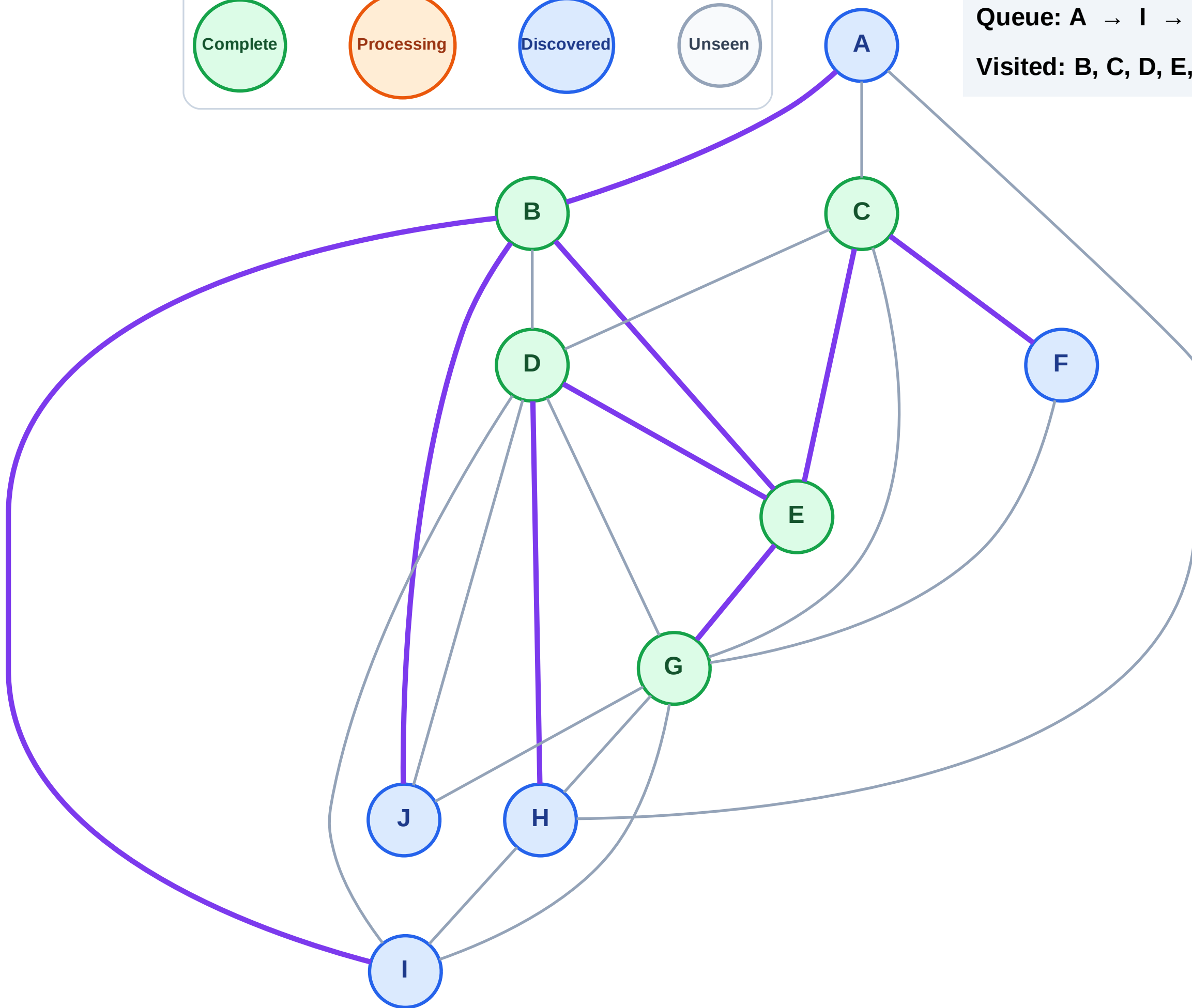
Step 11: No new neighbors from G

Legend



Queue: A → I → J → F → H

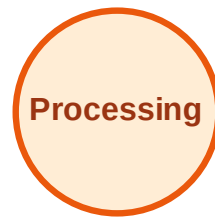
Visited: B, C, D, E, G



BFS Traversal

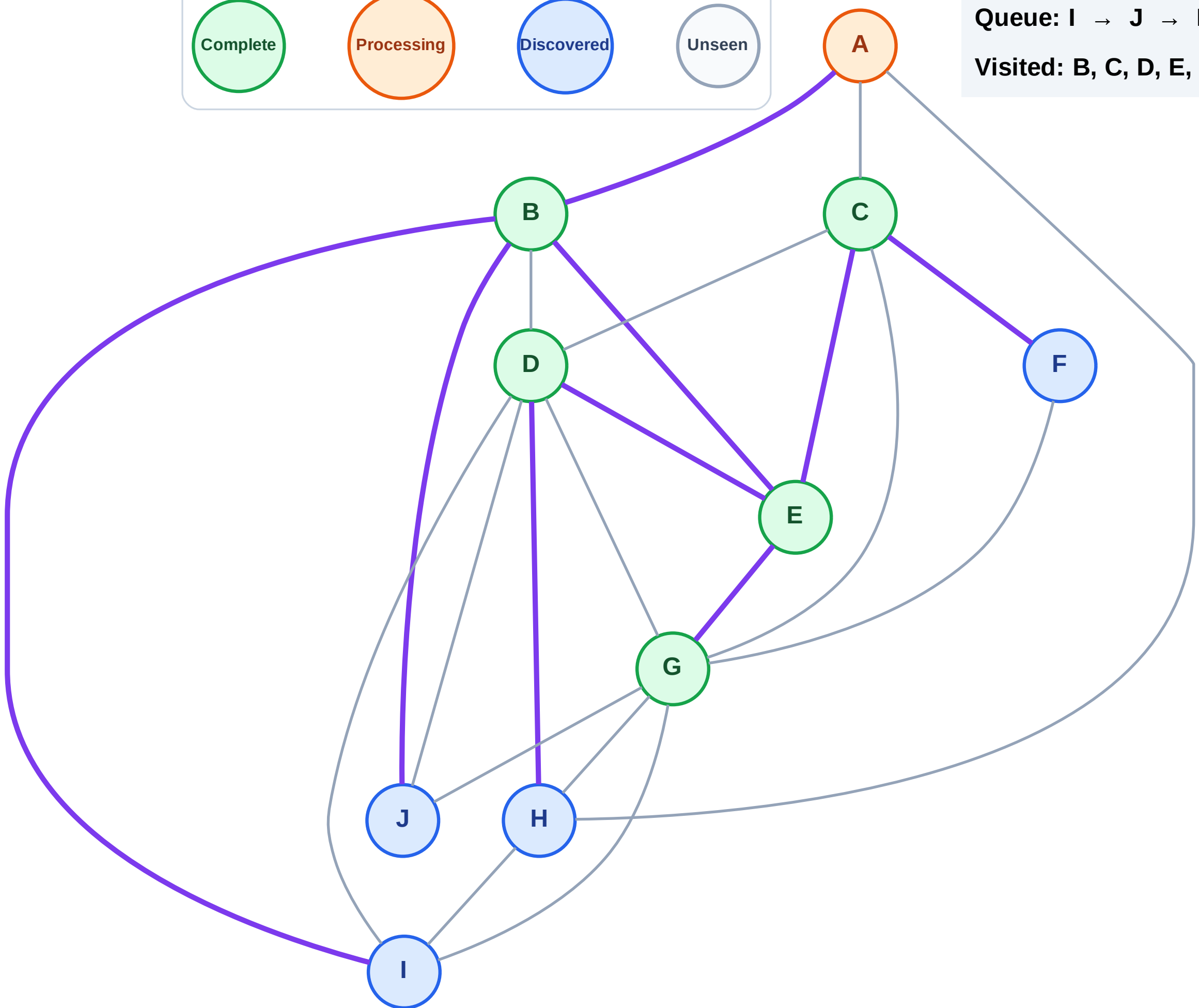
Step 12: Process node A

Legend



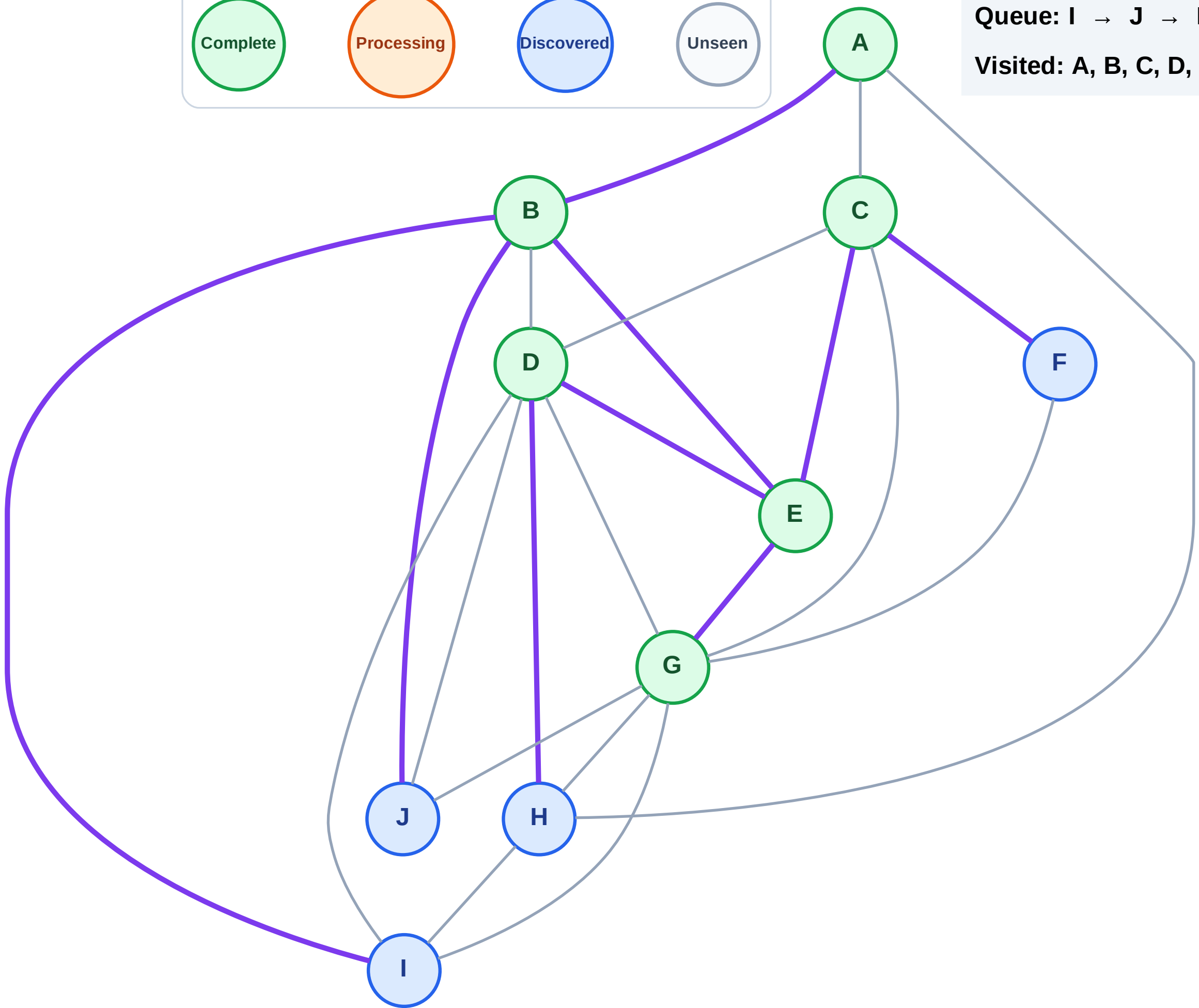
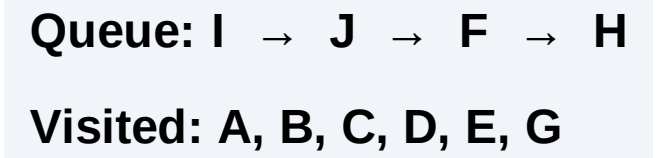
Queue: I → J → F → H

Visited: B, C, D, E, G



Step 13: No new neighbors from A

Legend



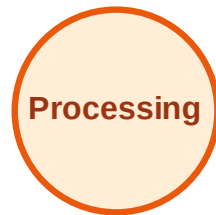
BFS Traversal

Step 14: Process node I

Legend



Complete



Processing

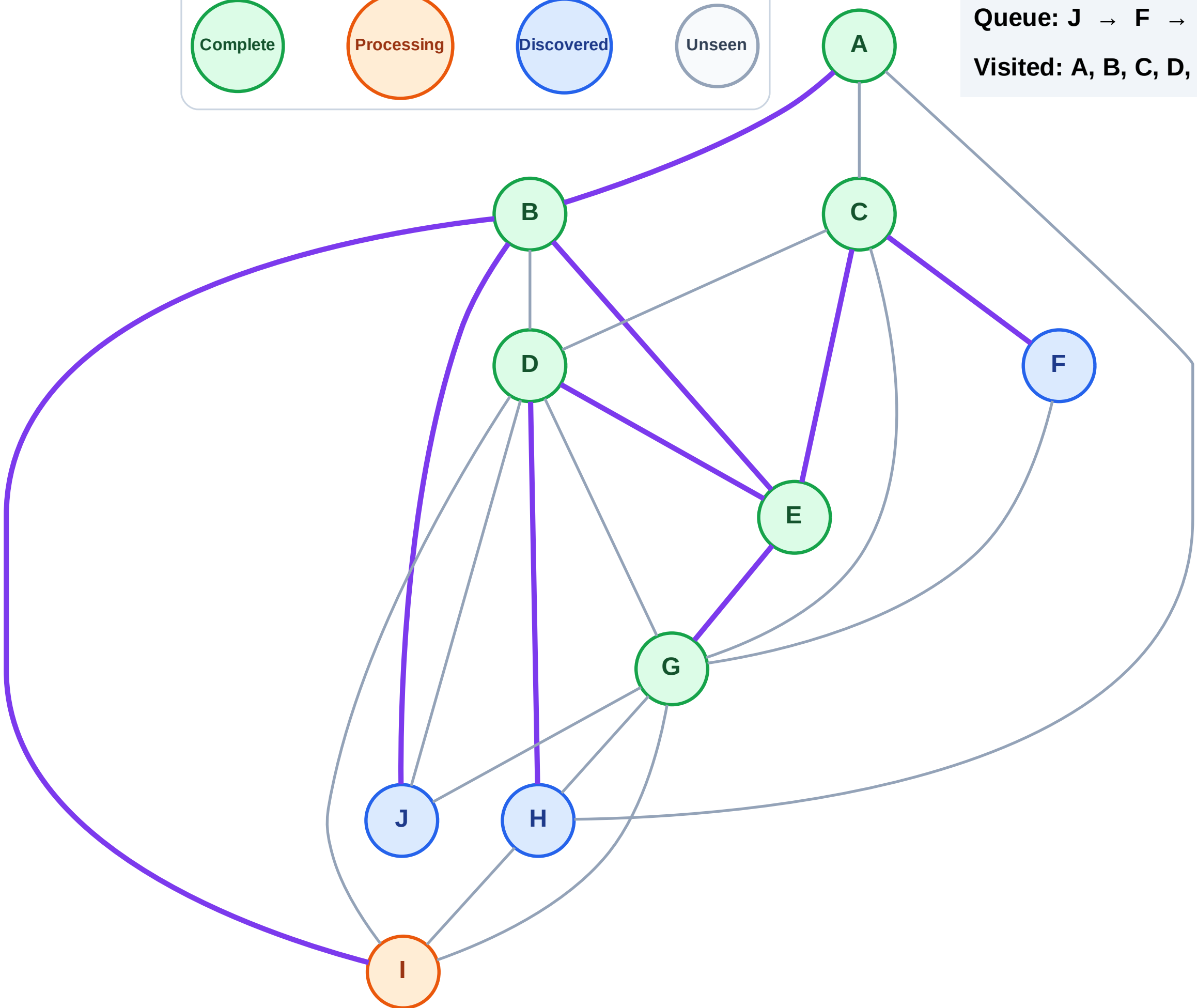


Discovered



Unseen

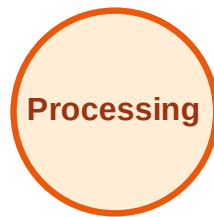
Queue: J → F → H
Visited: A, B, C, D, E, G



BFS Traversal

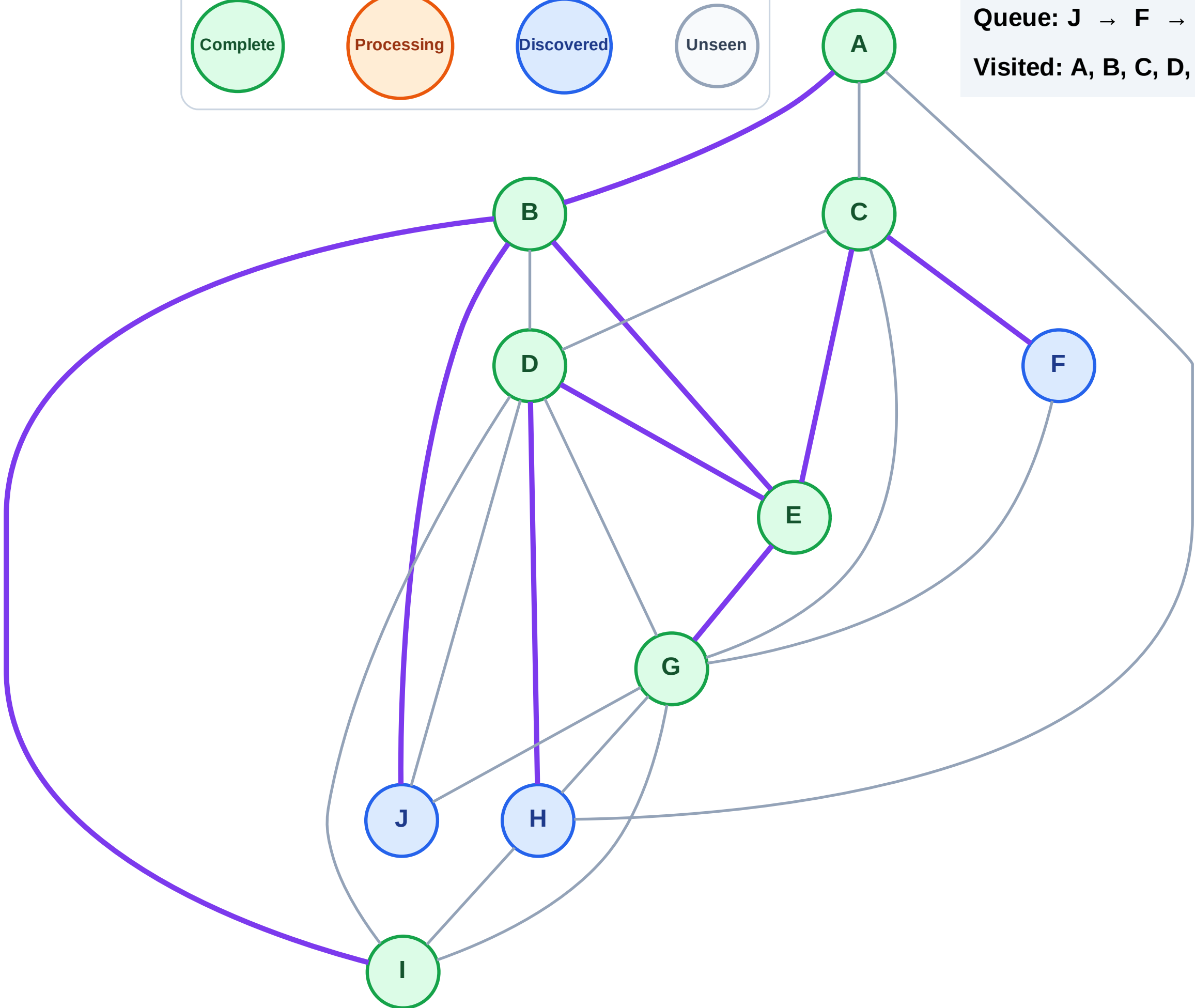
Step 15: No new neighbors from I

Legend



Queue: J → F → H

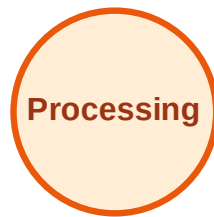
Visited: A, B, C, D, E, G, I



BFS Traversal

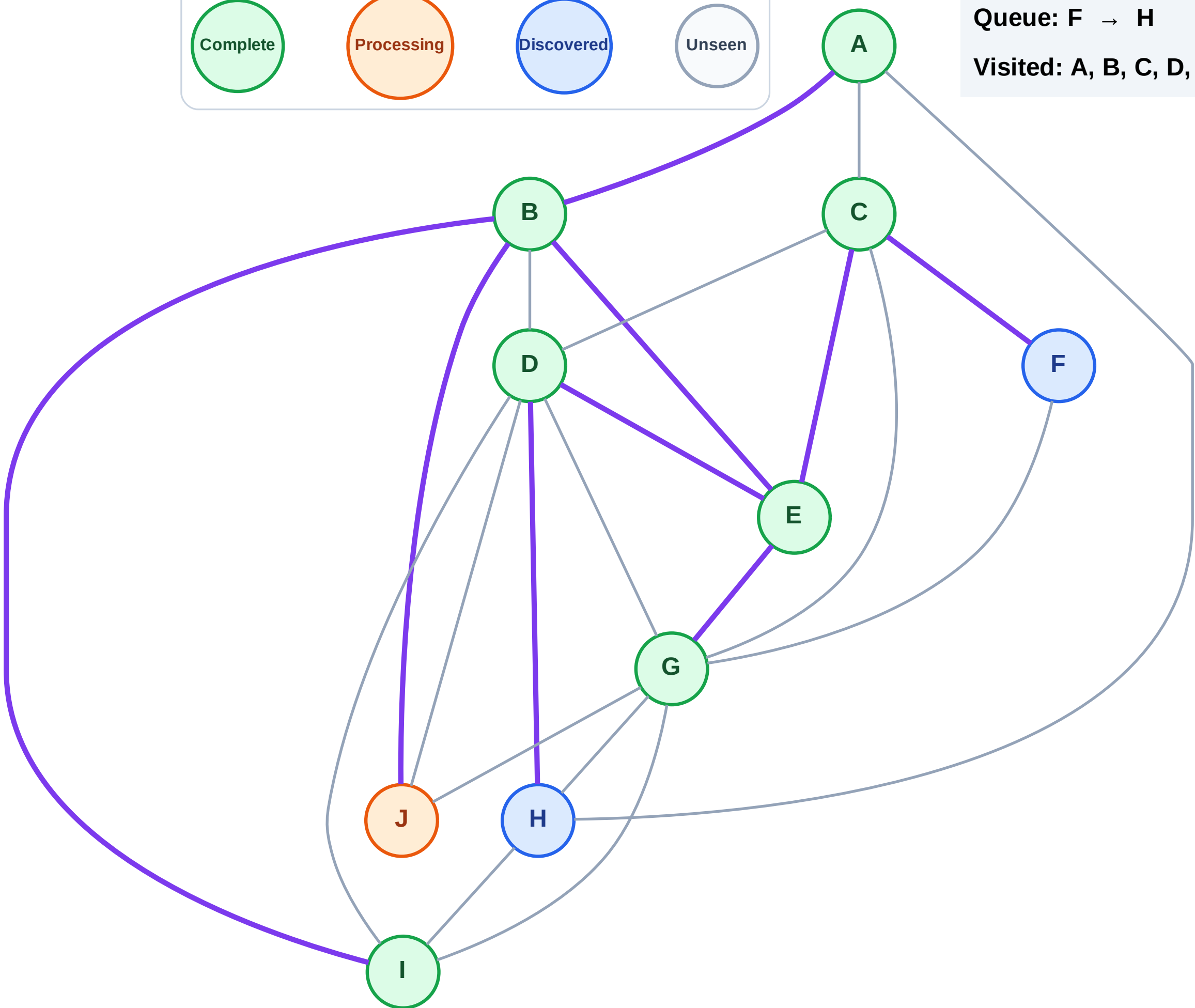
Step 16: Process node J

Legend



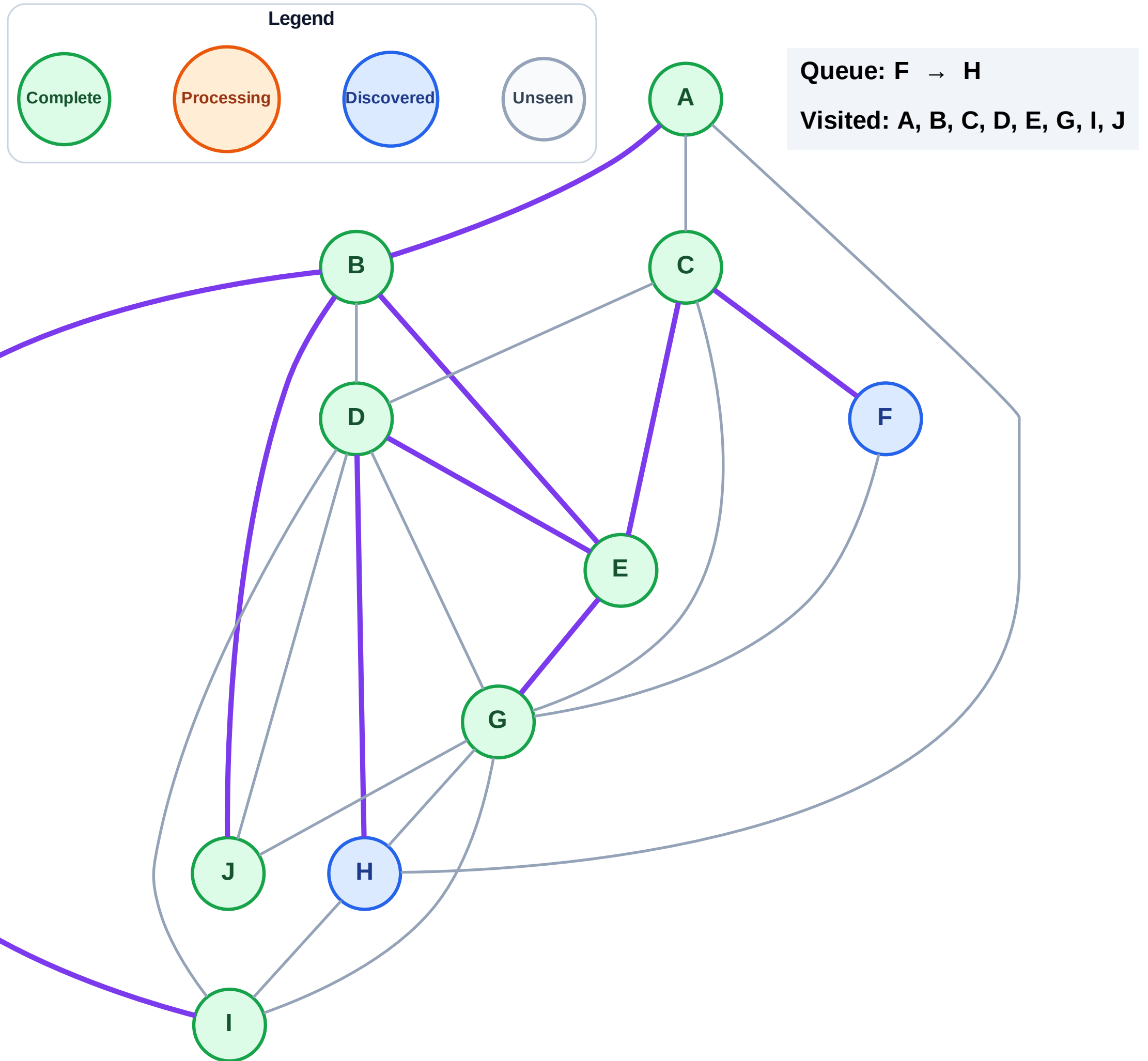
Queue: F → H

Visited: A, B, C, D, E, G, I



Step 17: No new neighbors from J

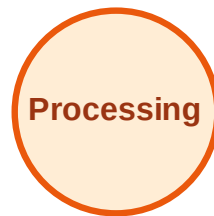
Step 17: No new neighbors from J



BFS Traversal

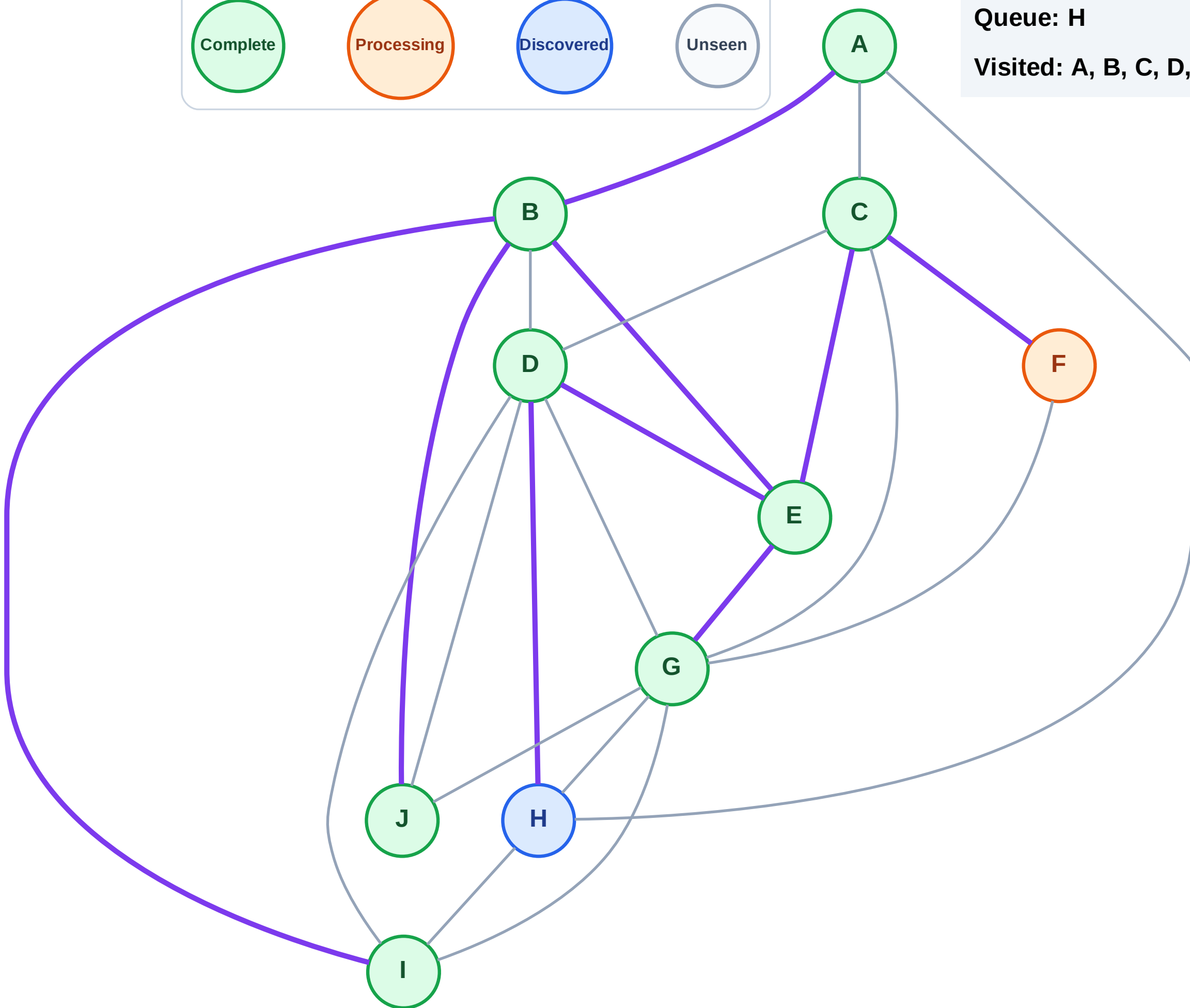
Step 18: Process node F

Legend



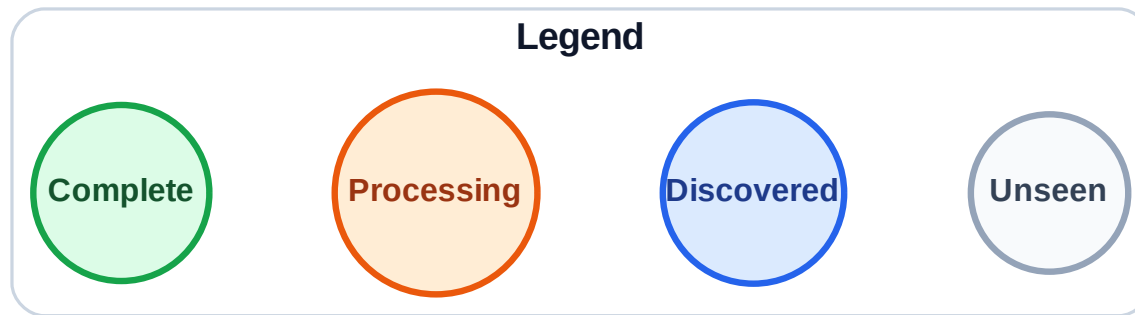
Queue: H

Visited: A, B, C, D, E, G, I, J



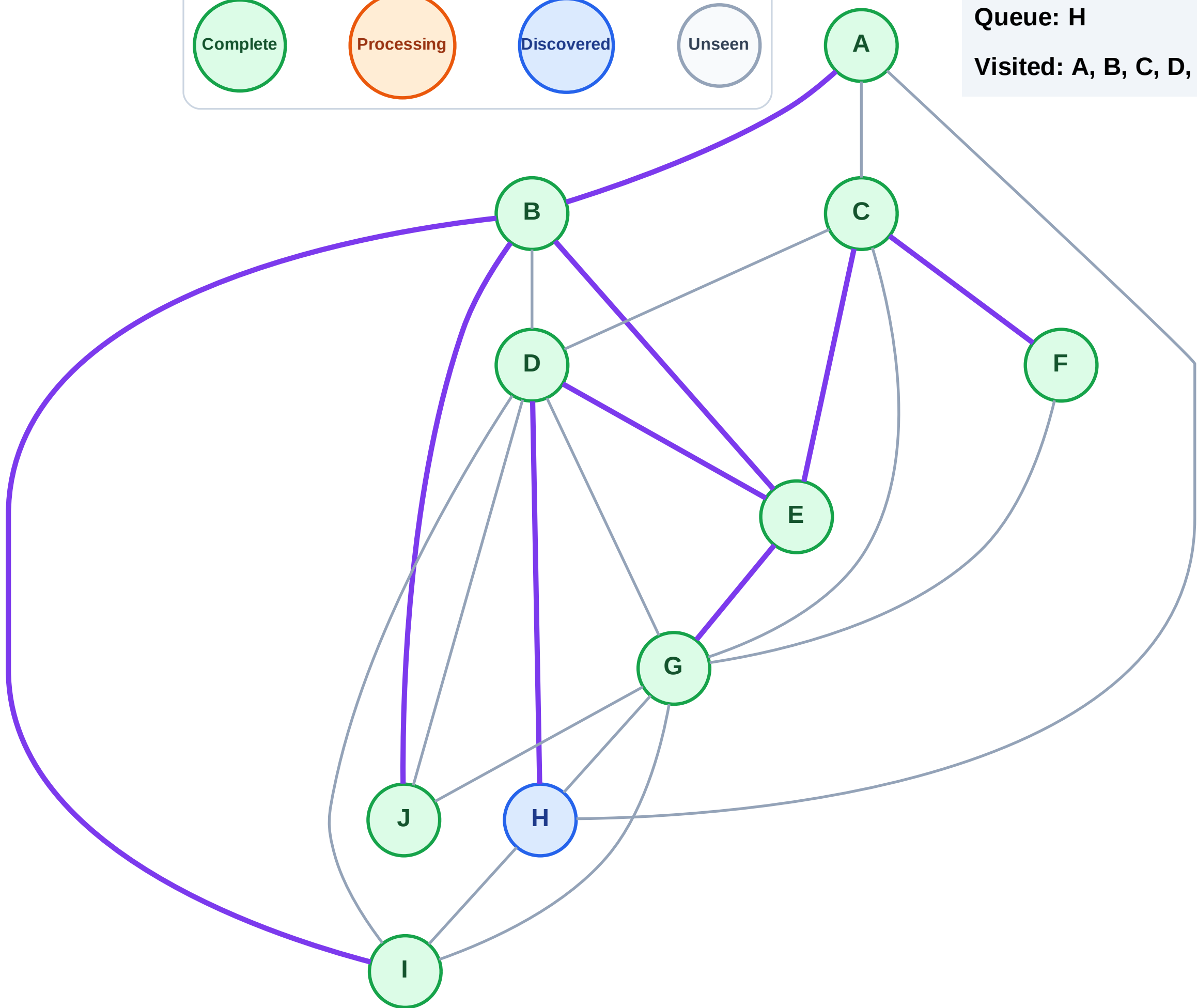
Step 19: No new neighbors from F

Step 19: No new neighbors from F



Queue: H

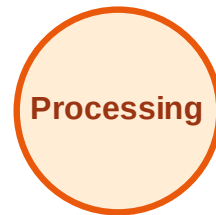
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

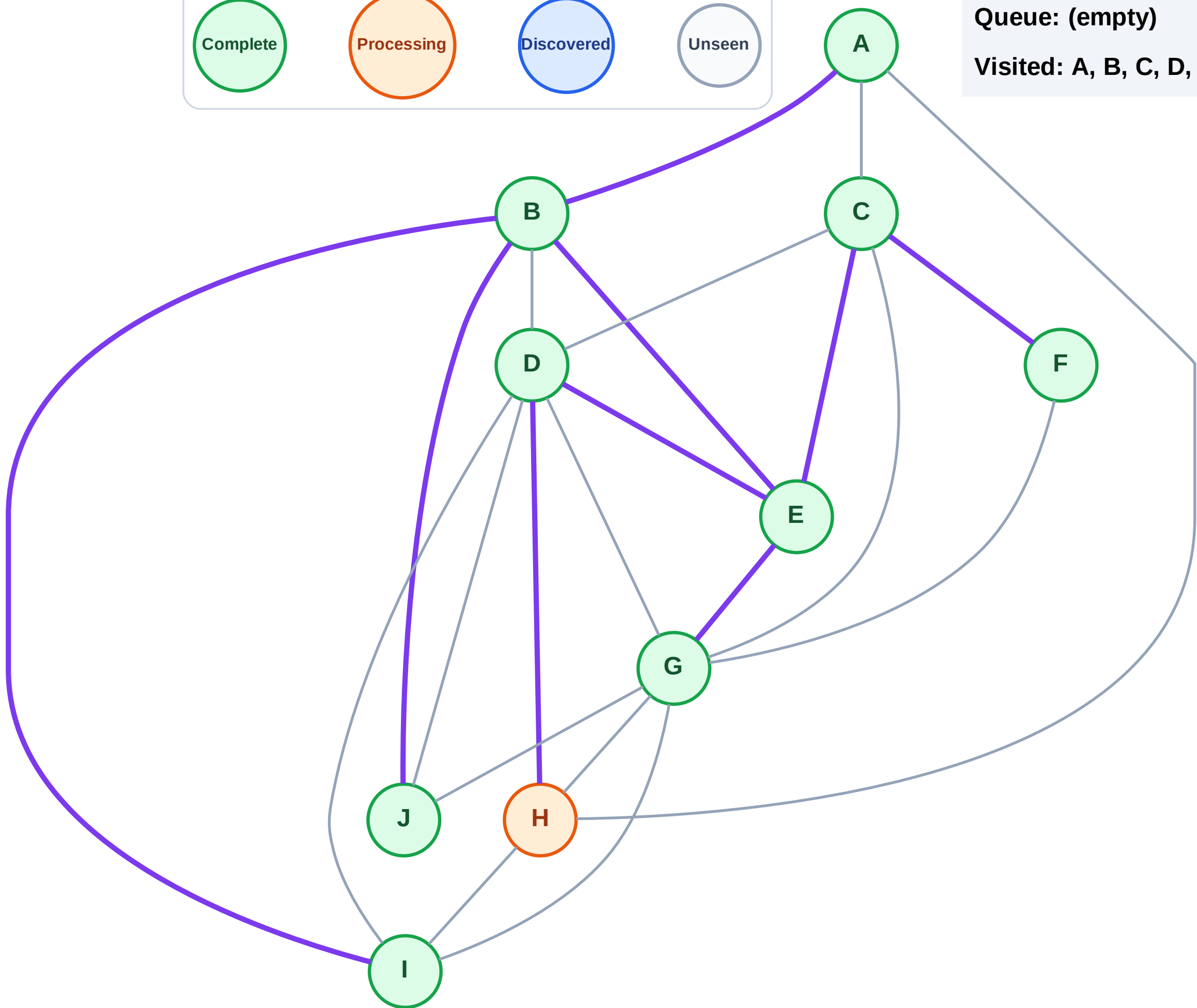
Step 20: Process node H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



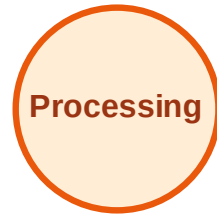
BFS Traversal

Step 21: No new neighbors from H

Legend



Complete



Processing



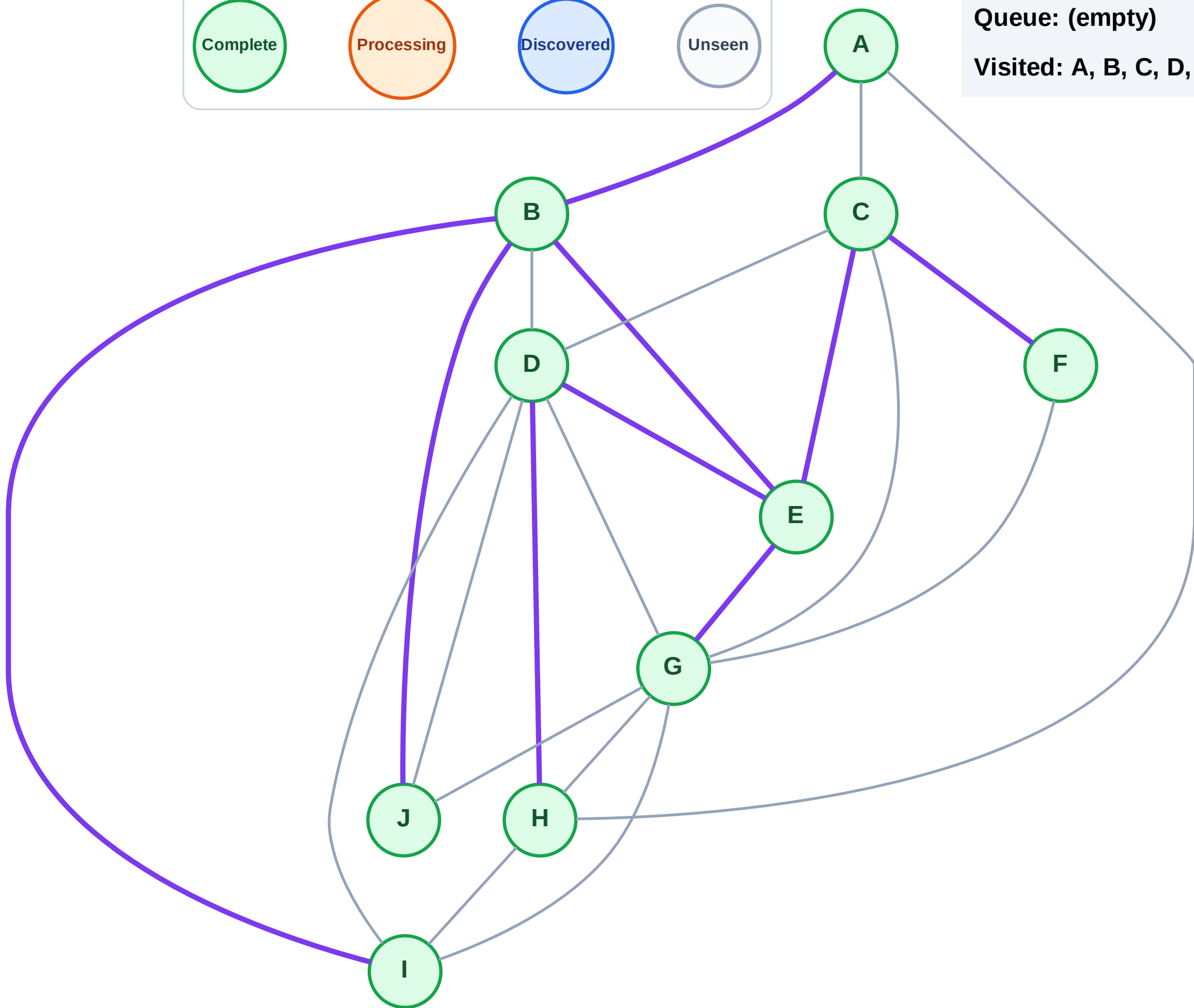
Discovered



Unseen

Queue: (empty)

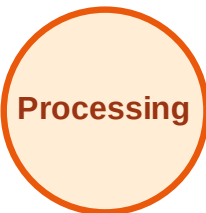
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

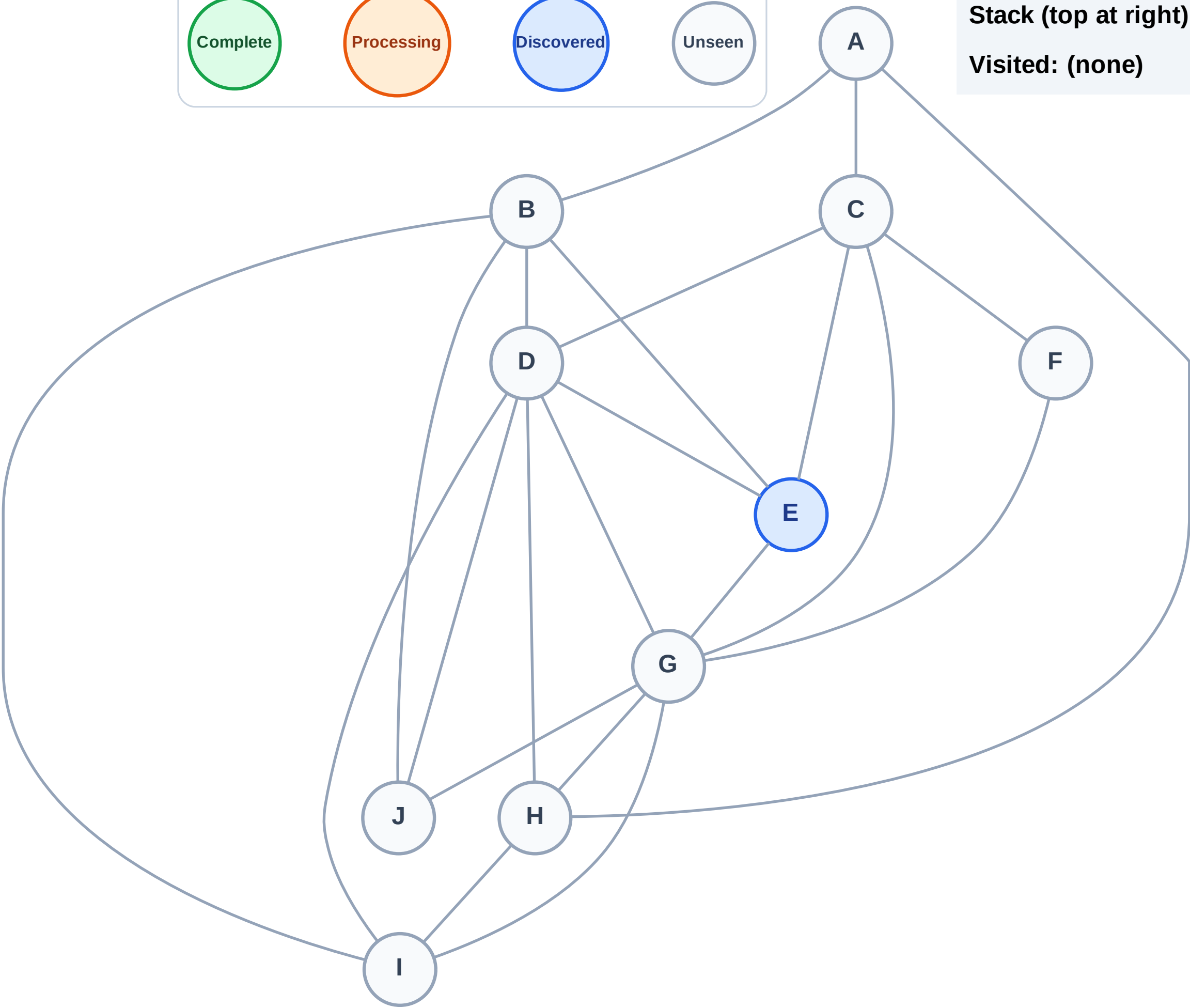
Step 1: Start at E

Legend



Stack (top at right): E

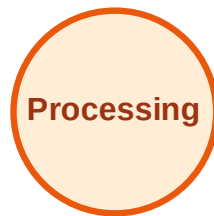
Visited: (none)



DFS Traversal

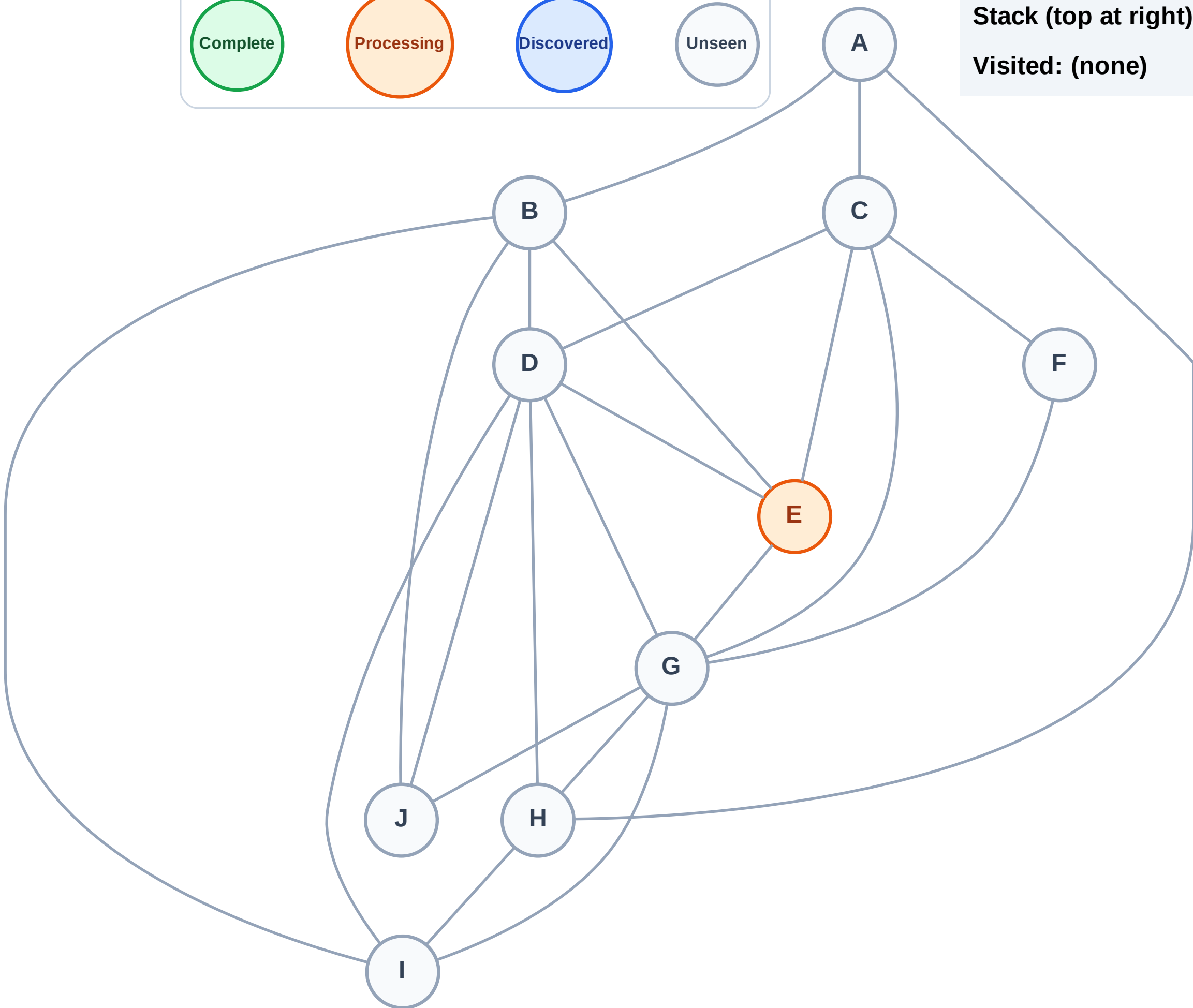
Step 2: Process node E

Legend



Stack (top at right): (empty)

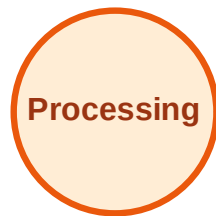
Visited: (none)



DFS Traversal

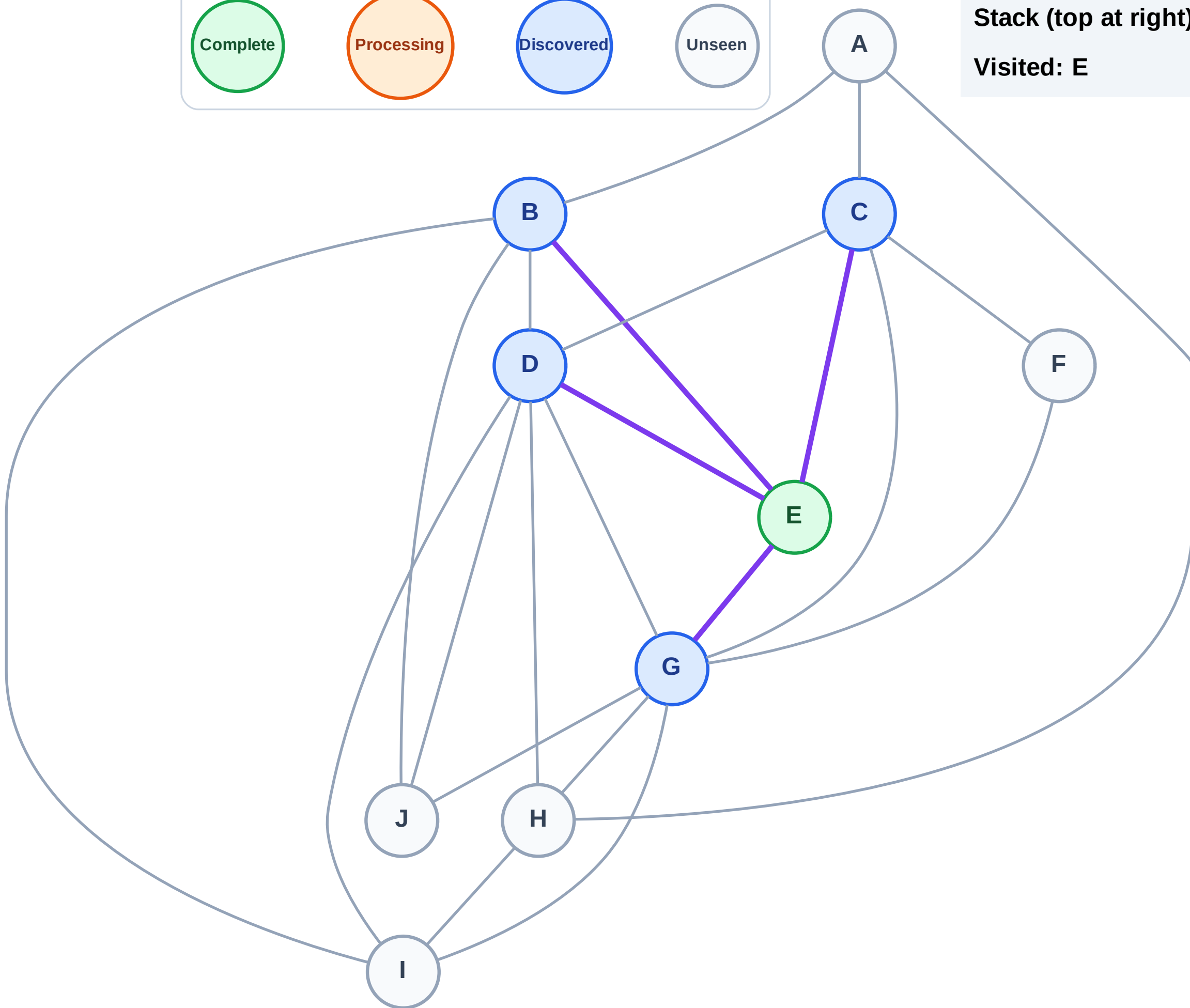
Step 3: Discover G, D, C, B from E

Legend



Stack (top at right): G | D | C | B

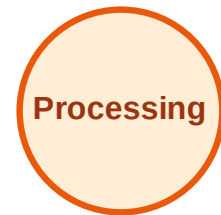
Visited: E



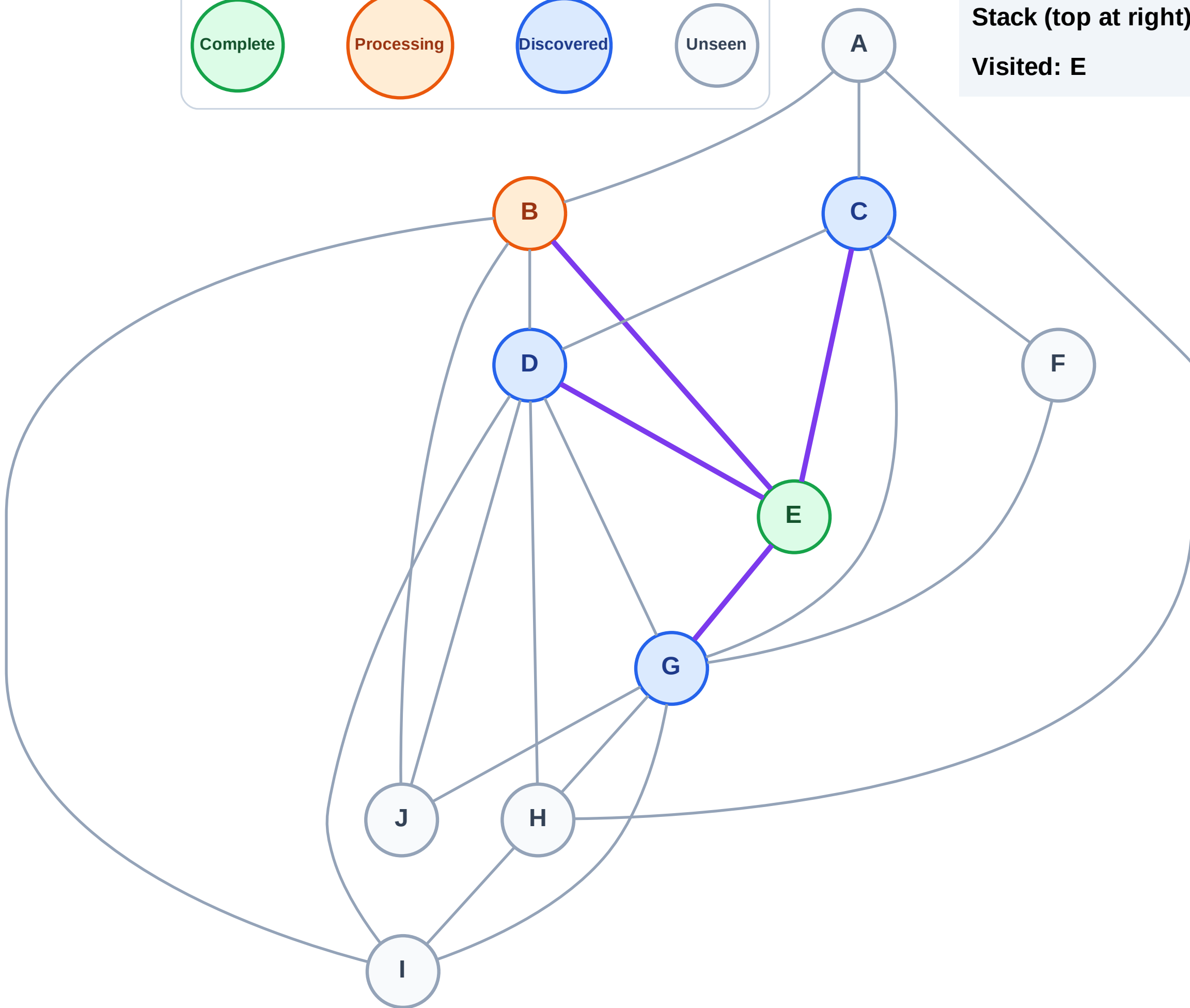
DFS Traversal

Step 4: Process node B

Legend



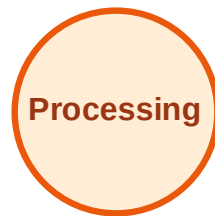
Stack (top at right): G | D | C
Visited: E



DFS Traversal

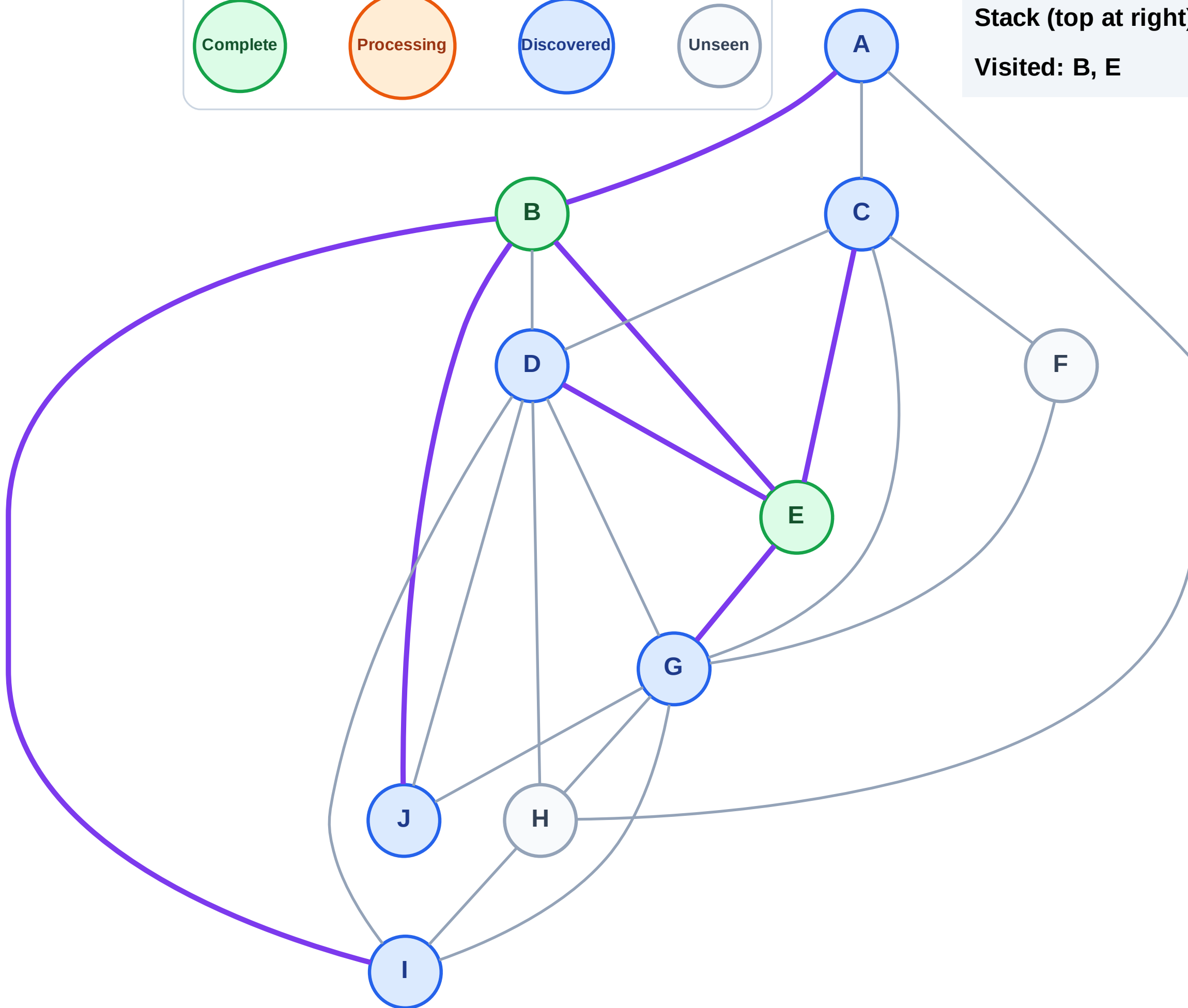
Step 5: Discover J, I, A from B

Legend



Stack (top at right): G | D | C | J | I | A

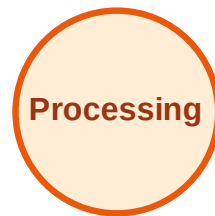
Visited: B, E



DFS Traversal

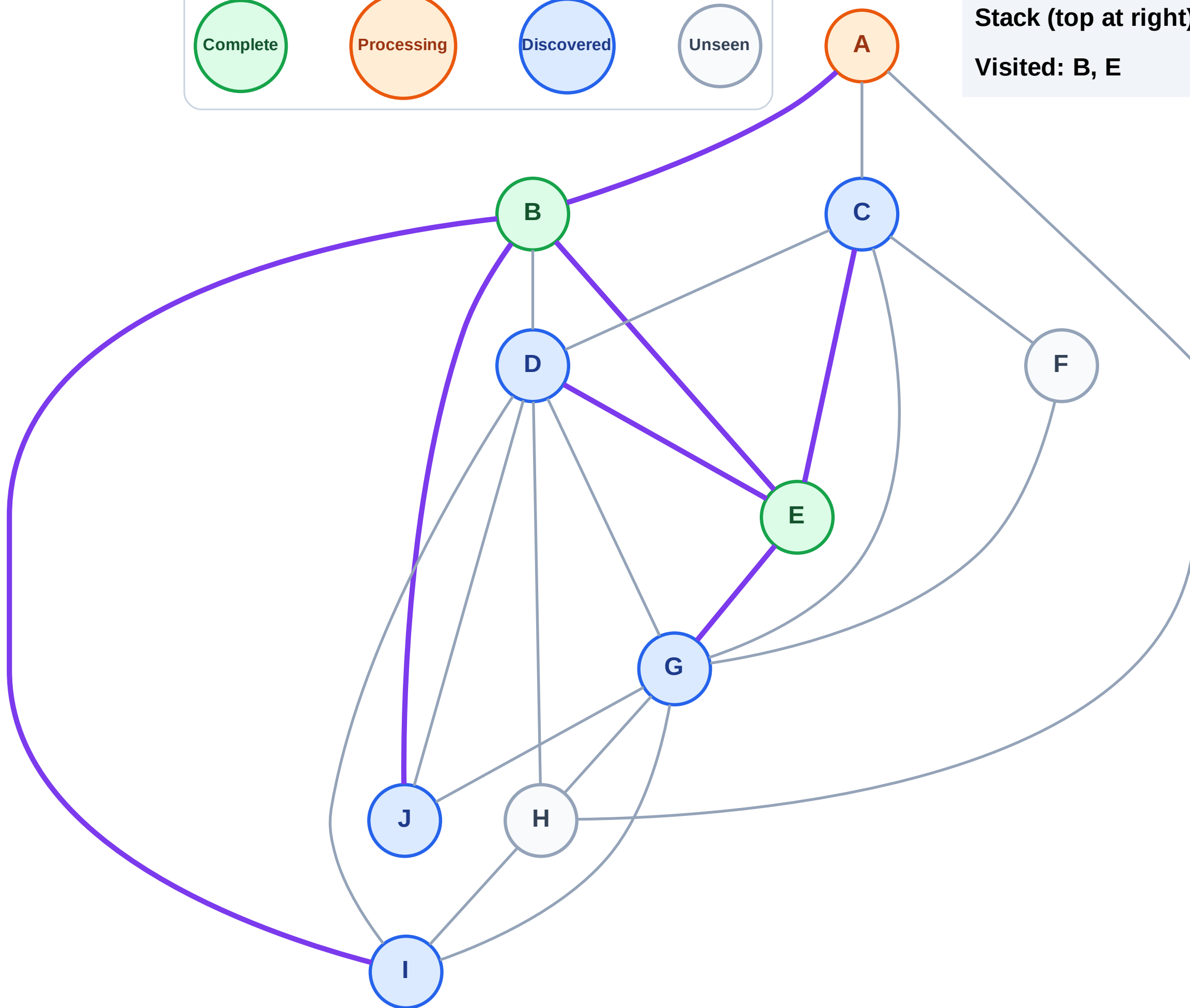
Step 6: Process node A

Legend



Stack (top at right): G | D | C | J | I

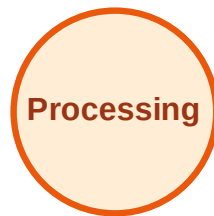
Visited: B, E



DFS Traversal

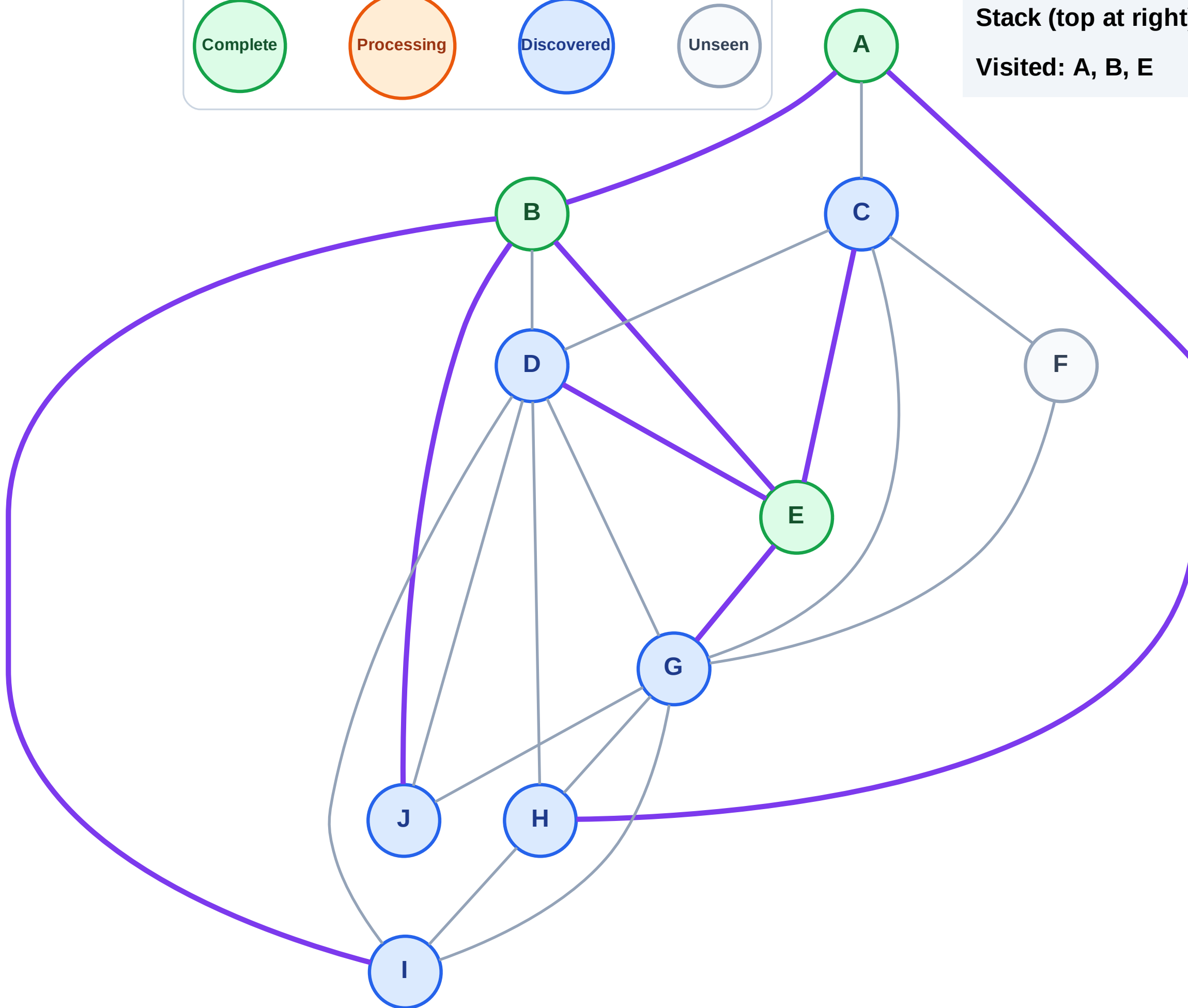
Step 7: Discover H from A

Legend



Stack (top at right): G | D | C | J | I | H

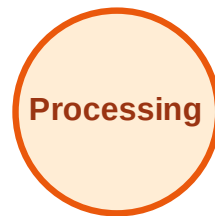
Visited: A, B, E



DFS Traversal

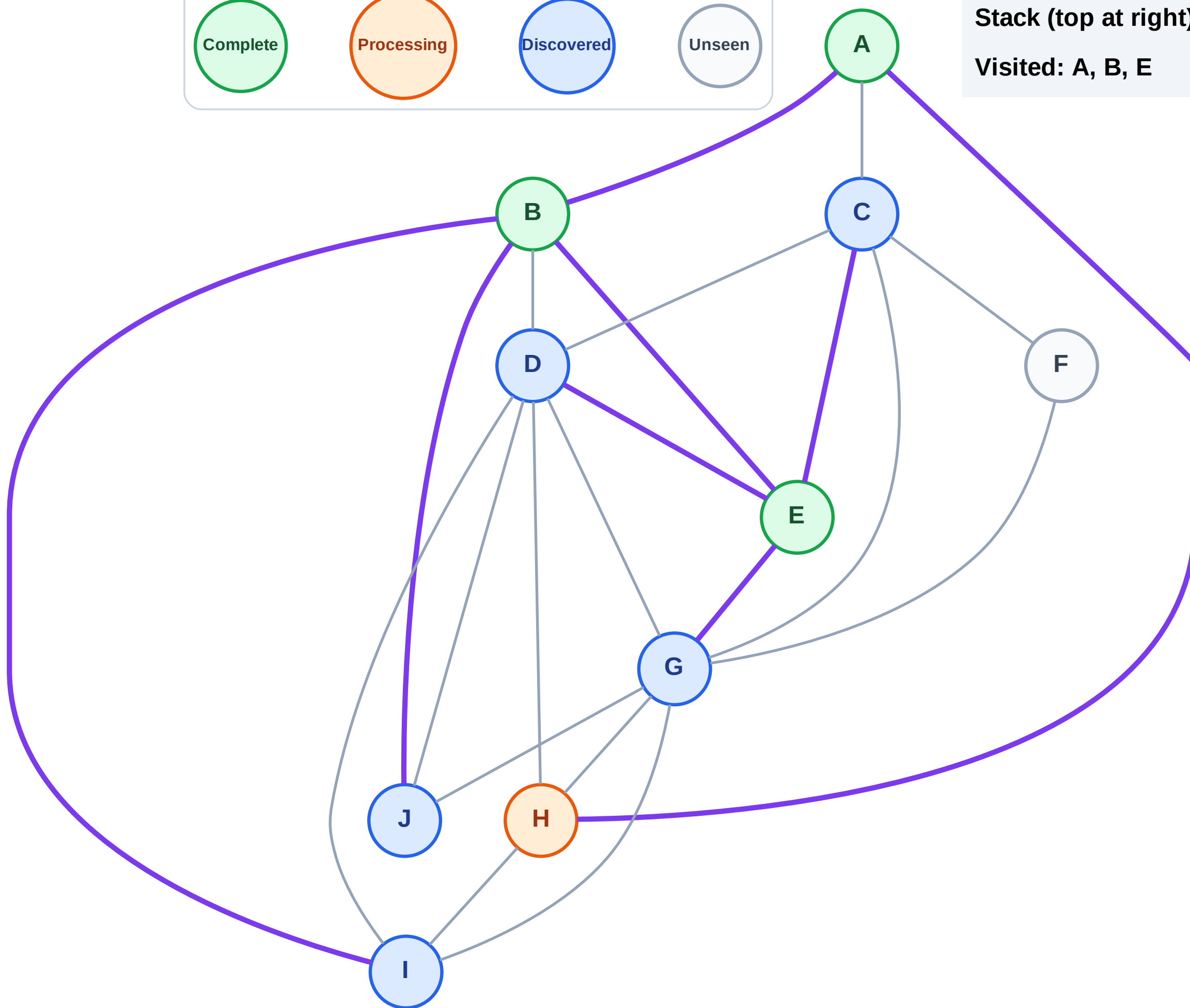
Step 8: Process node H

Legend



Stack (top at right): G | D | C | J | I

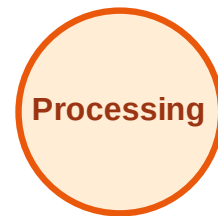
Visited: A, B, E



DFS Traversal

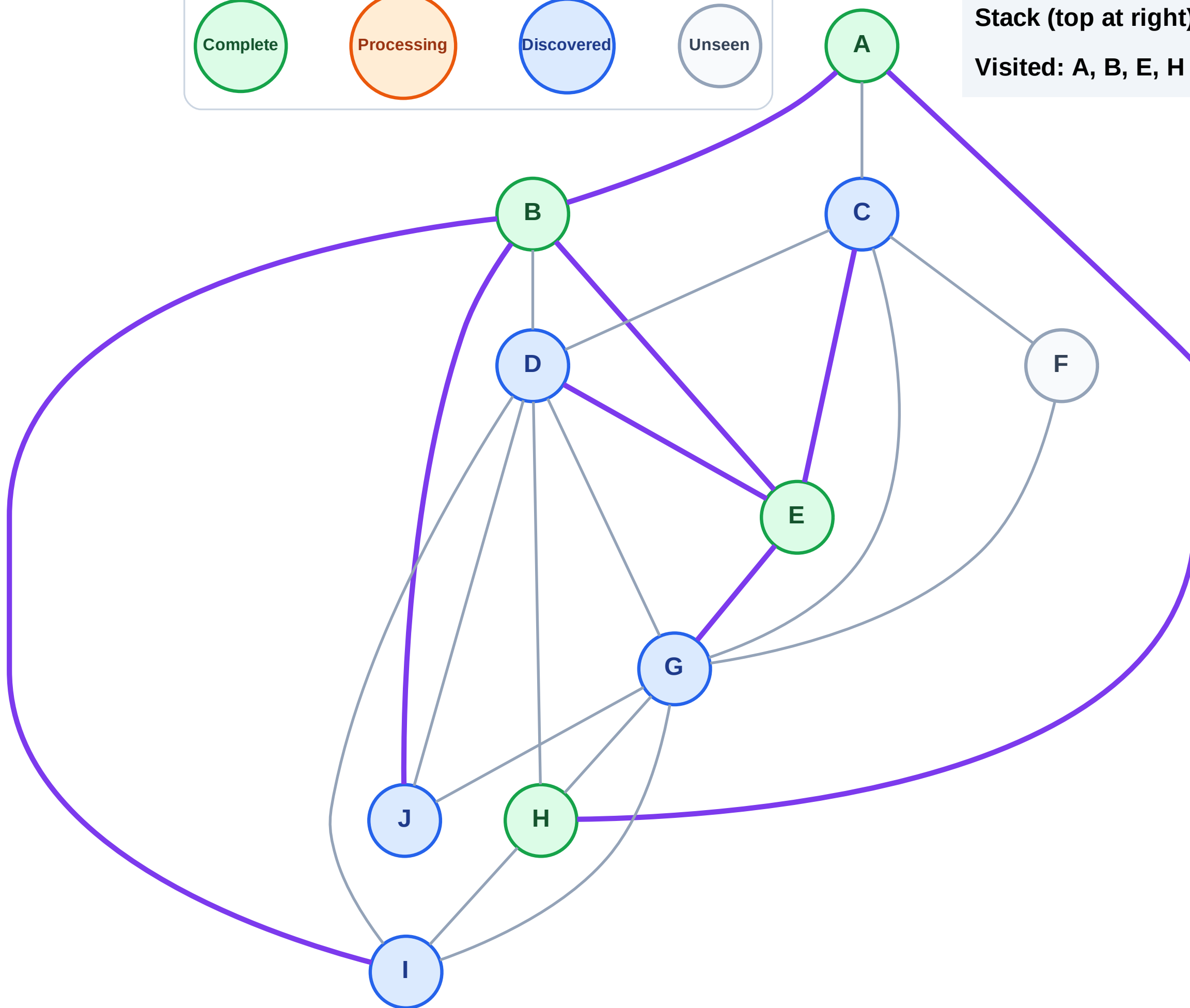
Step 9: No new neighbors from H

Legend



Stack (top at right): G | D | C | J | I

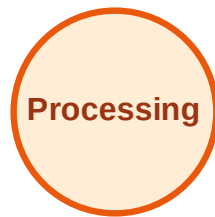
Visited: A, B, E, H



DFS Traversal

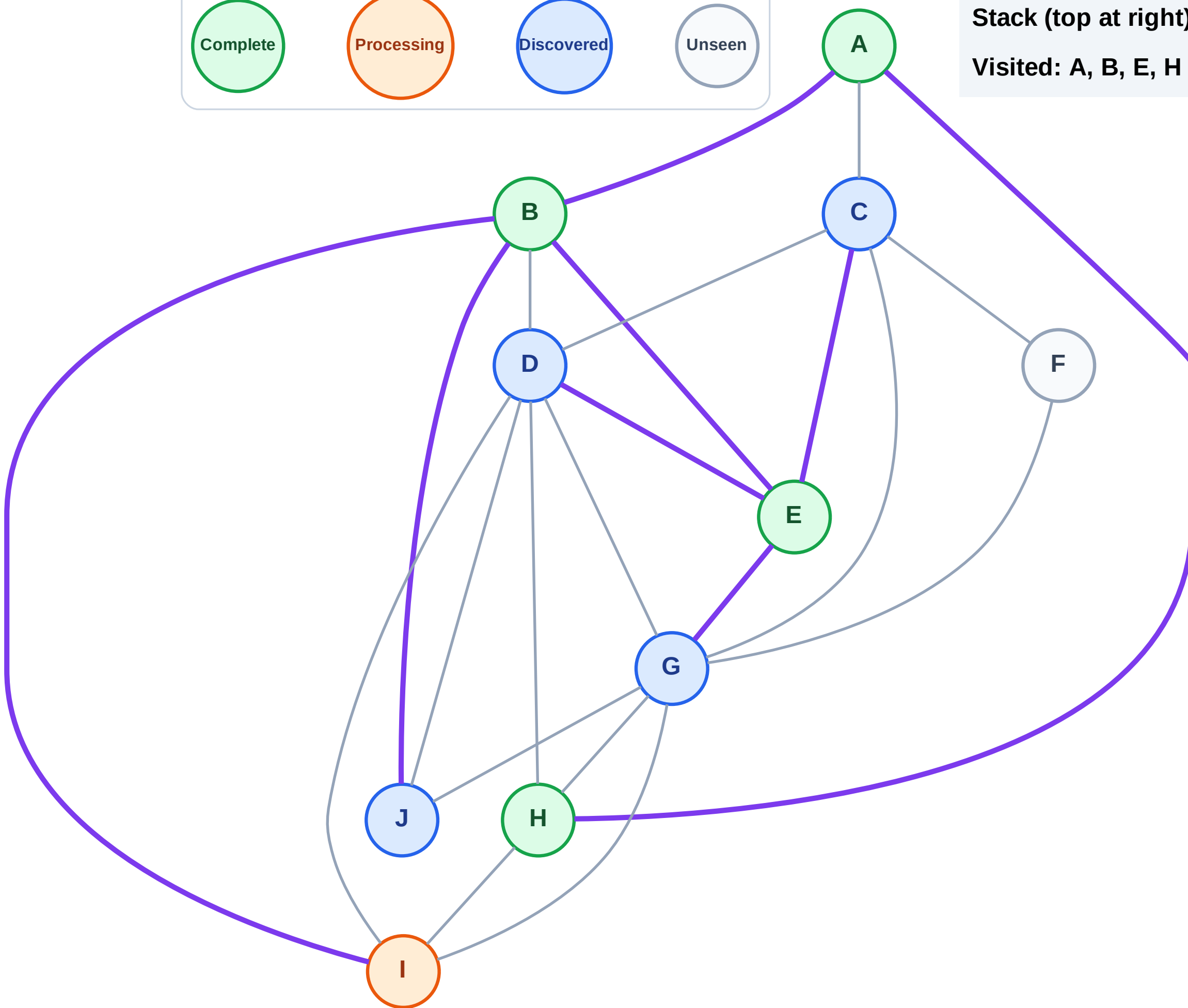
Step 10: Process node I

Legend



Stack (top at right): G | D | C | J

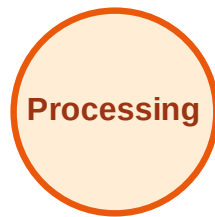
Visited: A, B, E, H



DFS Traversal

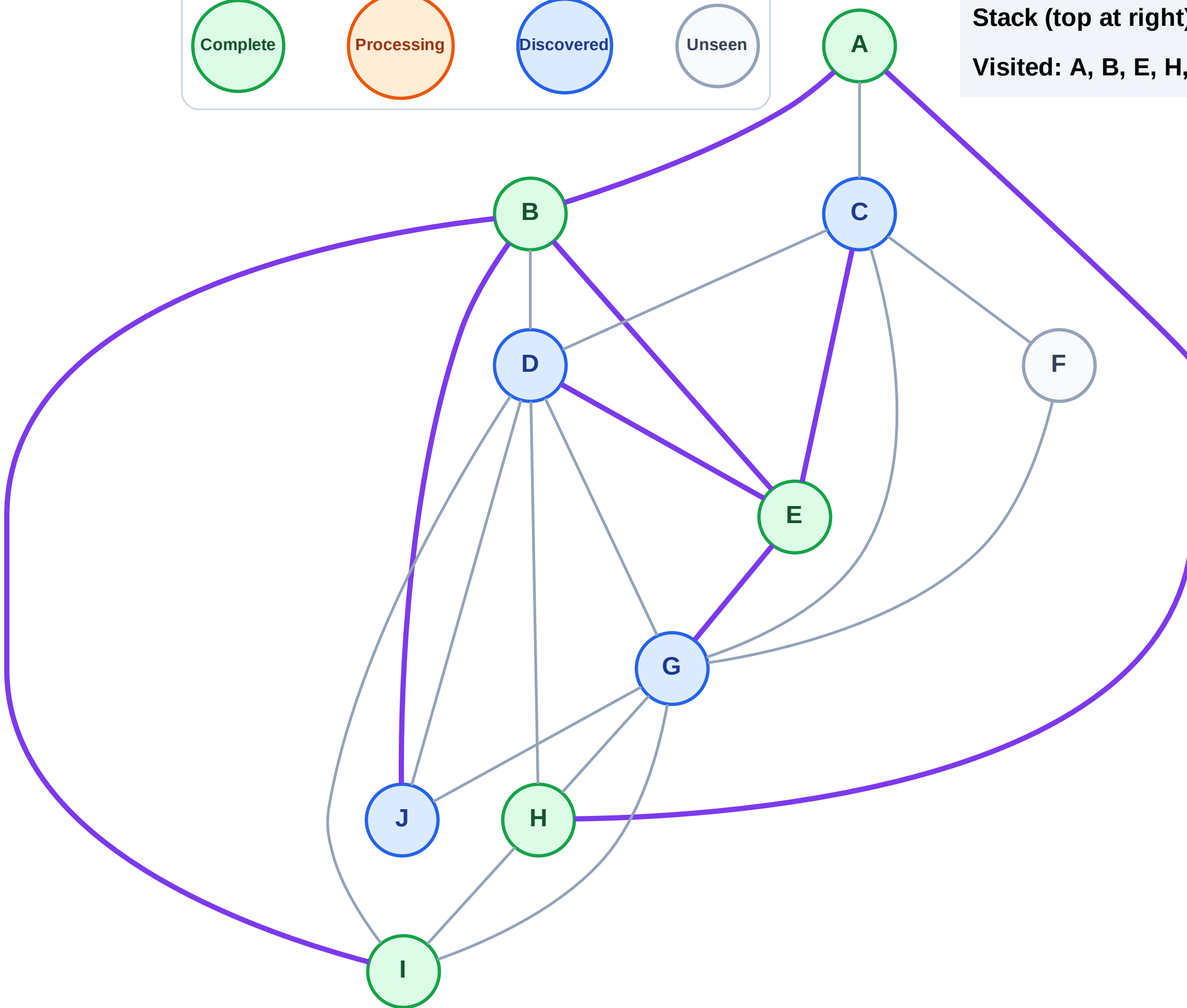
Step 11: No new neighbors from I

Legend



Stack (top at right): G | D | C | J

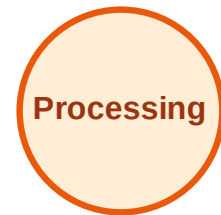
Visited: A, B, E, H, I



DFS Traversal

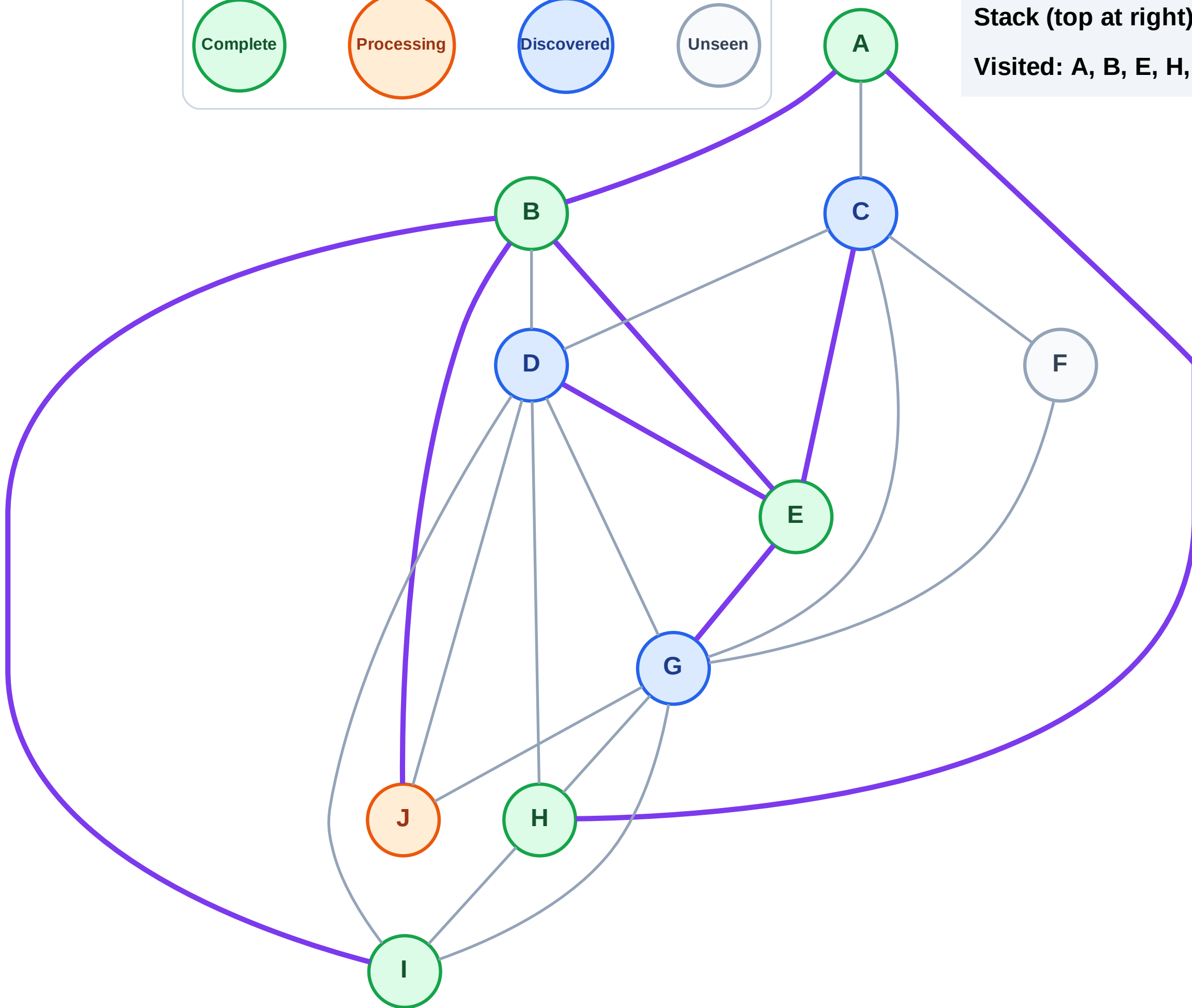
Step 12: Process node J

Legend



Stack (top at right): G | D | C

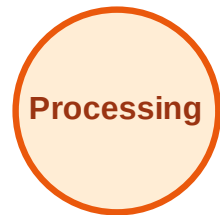
Visited: A, B, E, H, I



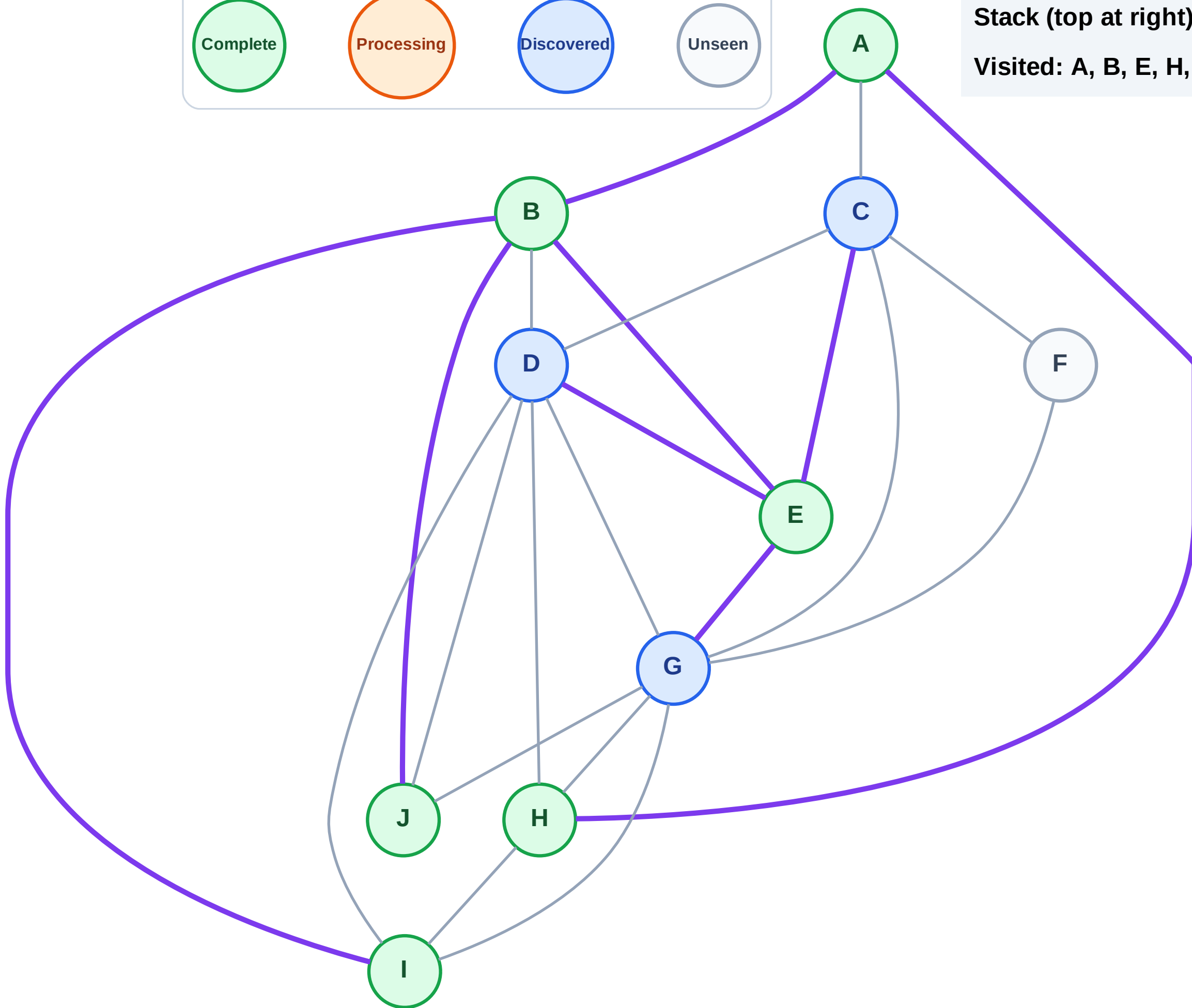
DFS Traversal

Step 13: No new neighbors from J

Legend

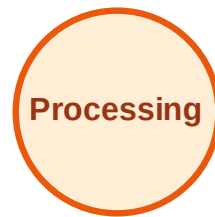


Stack (top at right): G | D | C
Visited: A, B, E, H, I, J



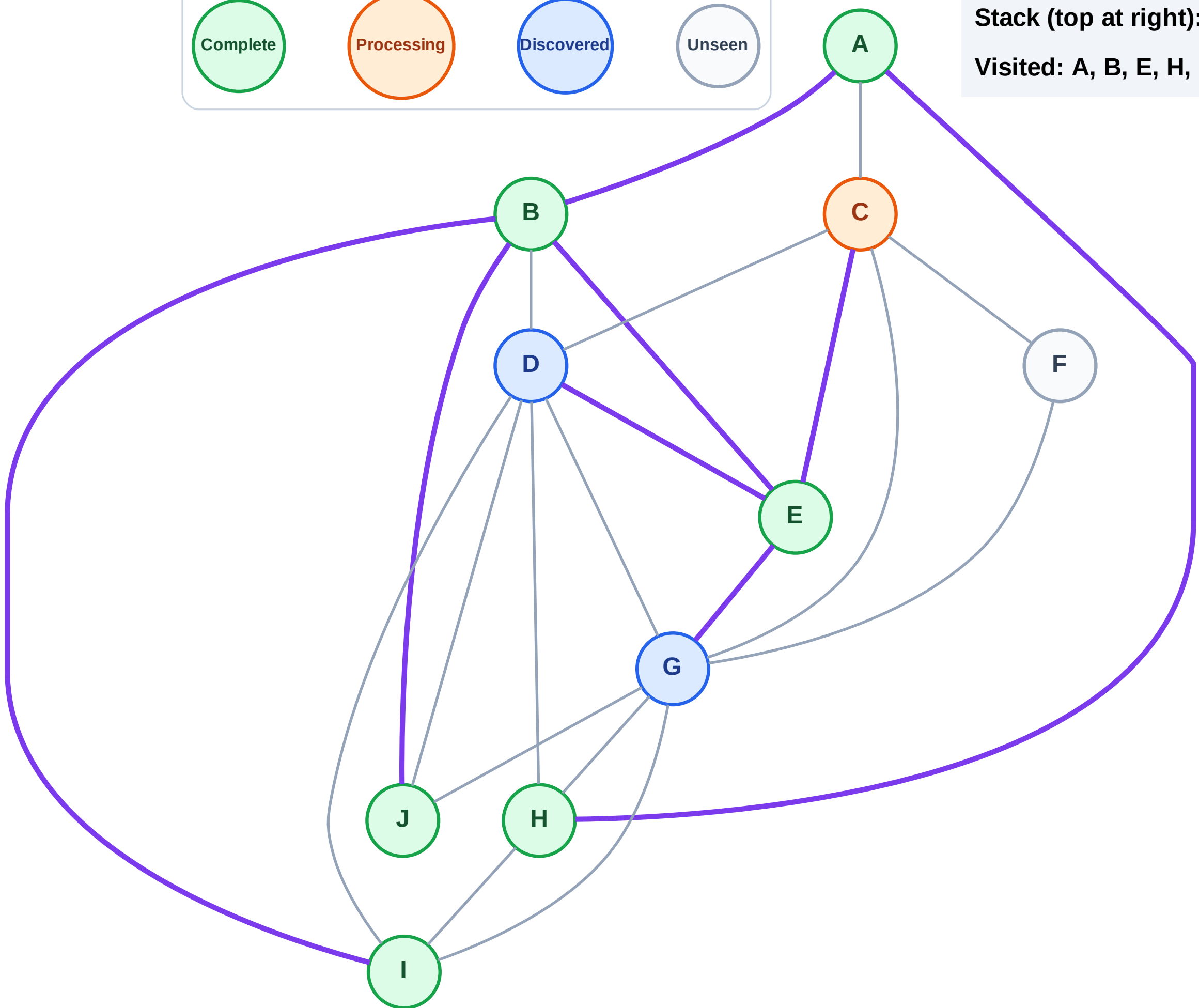
Step 14: Process node C

Legend



Stack (top at right): G | D

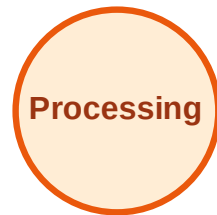
Visited: A, B, E, H, I, J



DFS Traversal

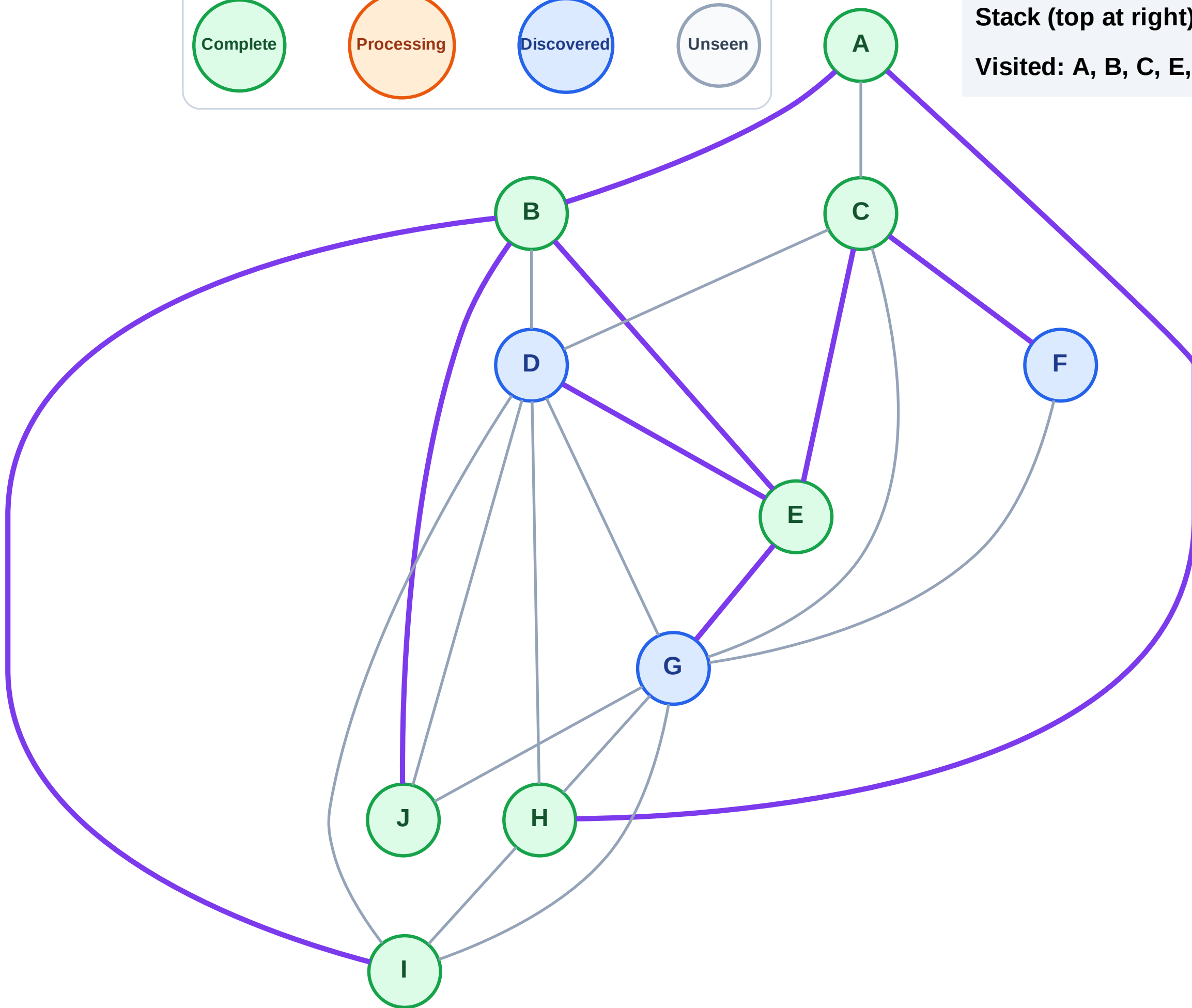
Step 15: Discover F from C

Legend



Stack (top at right): G | D | F

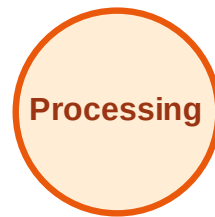
Visited: A, B, C, E, H, I, J



DFS Traversal

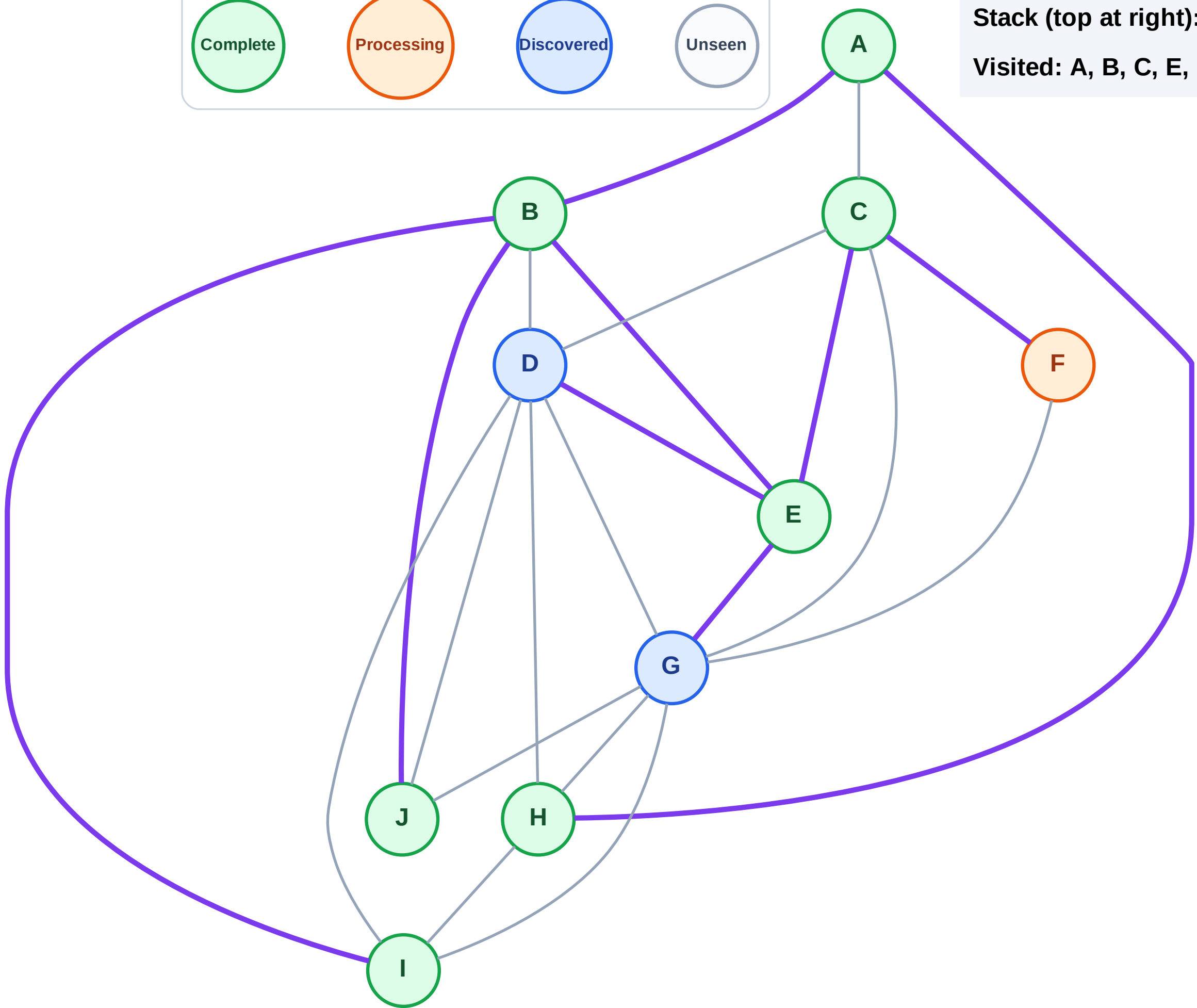
Step 16: Process node F

Legend



Stack (top at right): G | D

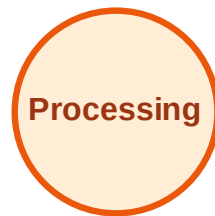
Visited: A, B, C, E, H, I, J



DFS Traversal

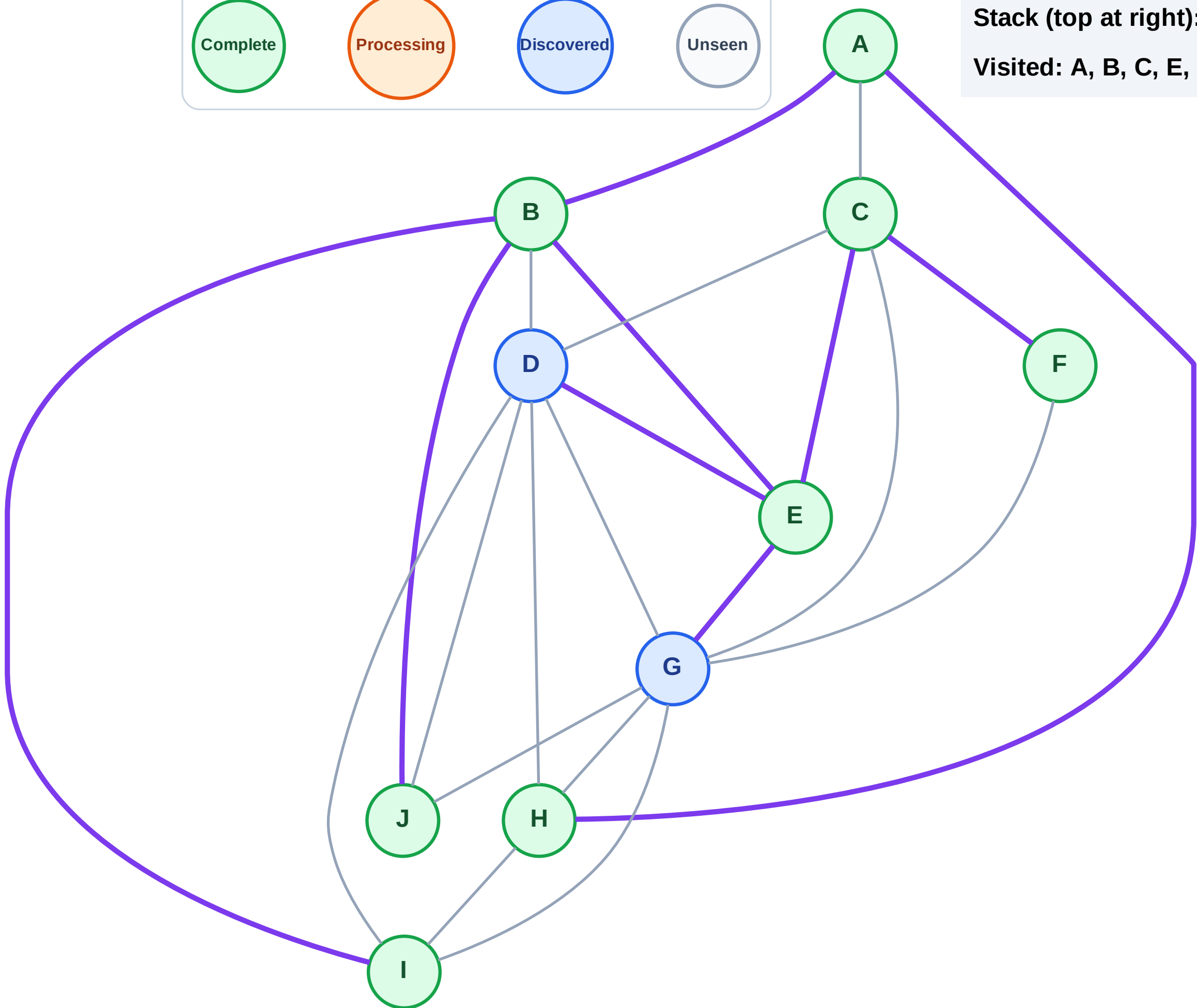
Step 17: No new neighbors from F

Legend



Stack (top at right): G | D

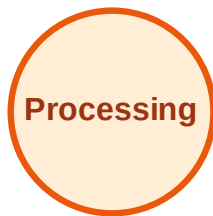
Visited: A, B, C, E, F, H, I, J



DFS Traversal

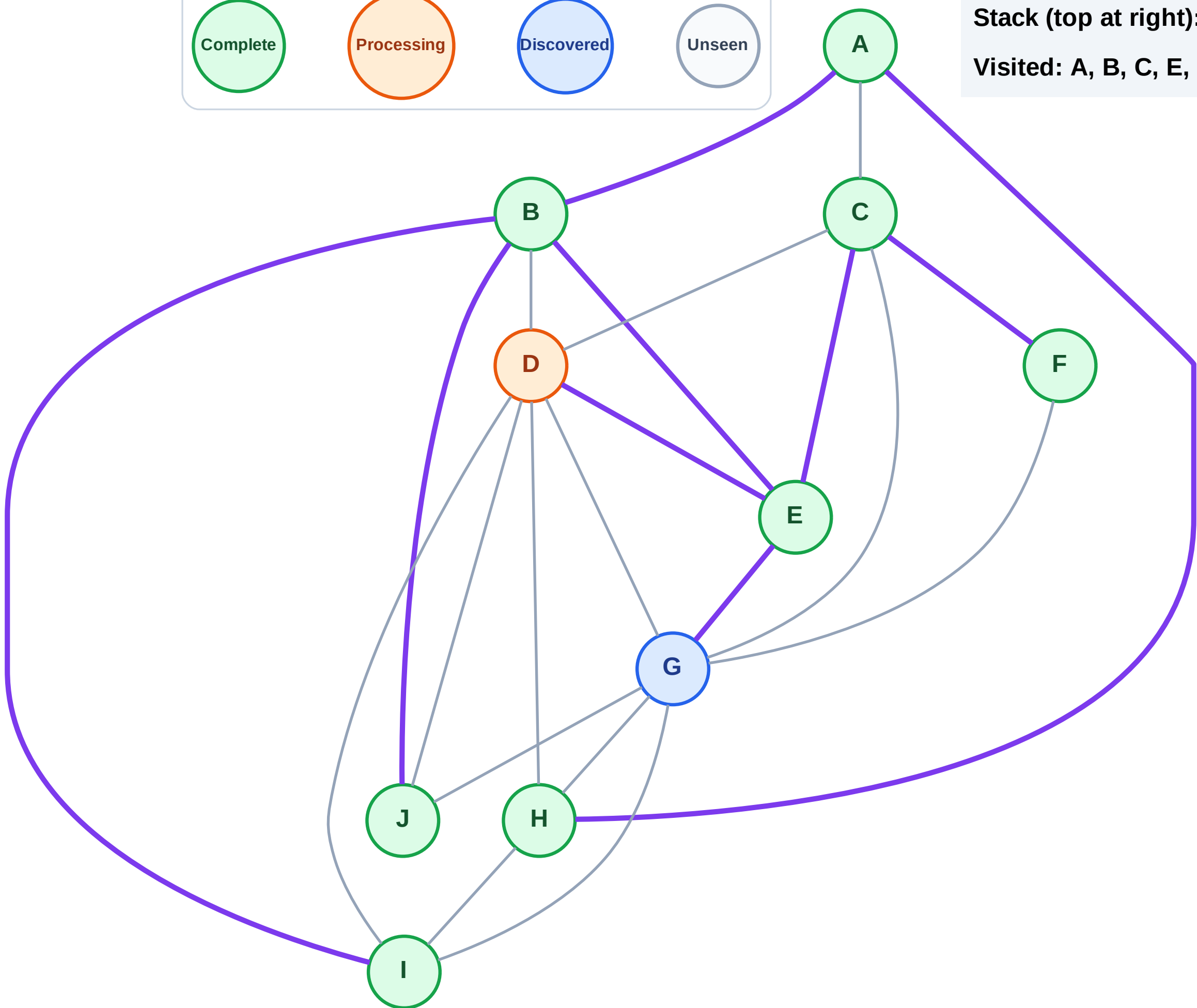
Step 18: Process node D

Legend



Stack (top at right): G

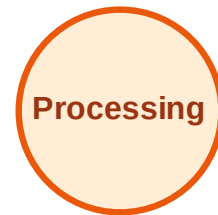
Visited: A, B, C, E, F, H, I, J



DFS Traversal

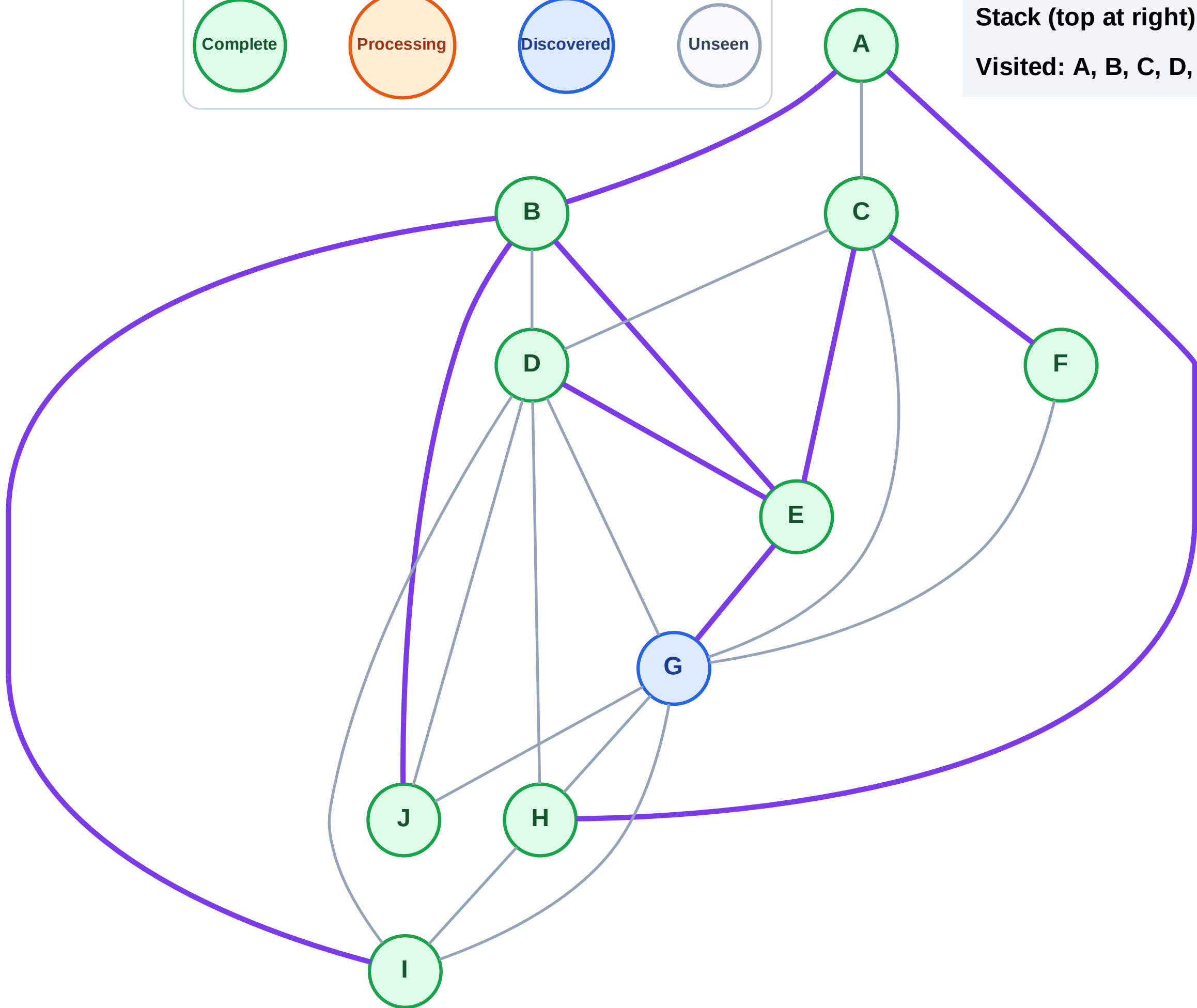
Step 19: No new neighbors from D

Legend



Stack (top at right): G

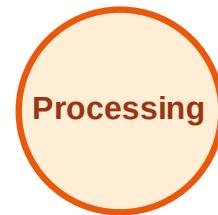
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

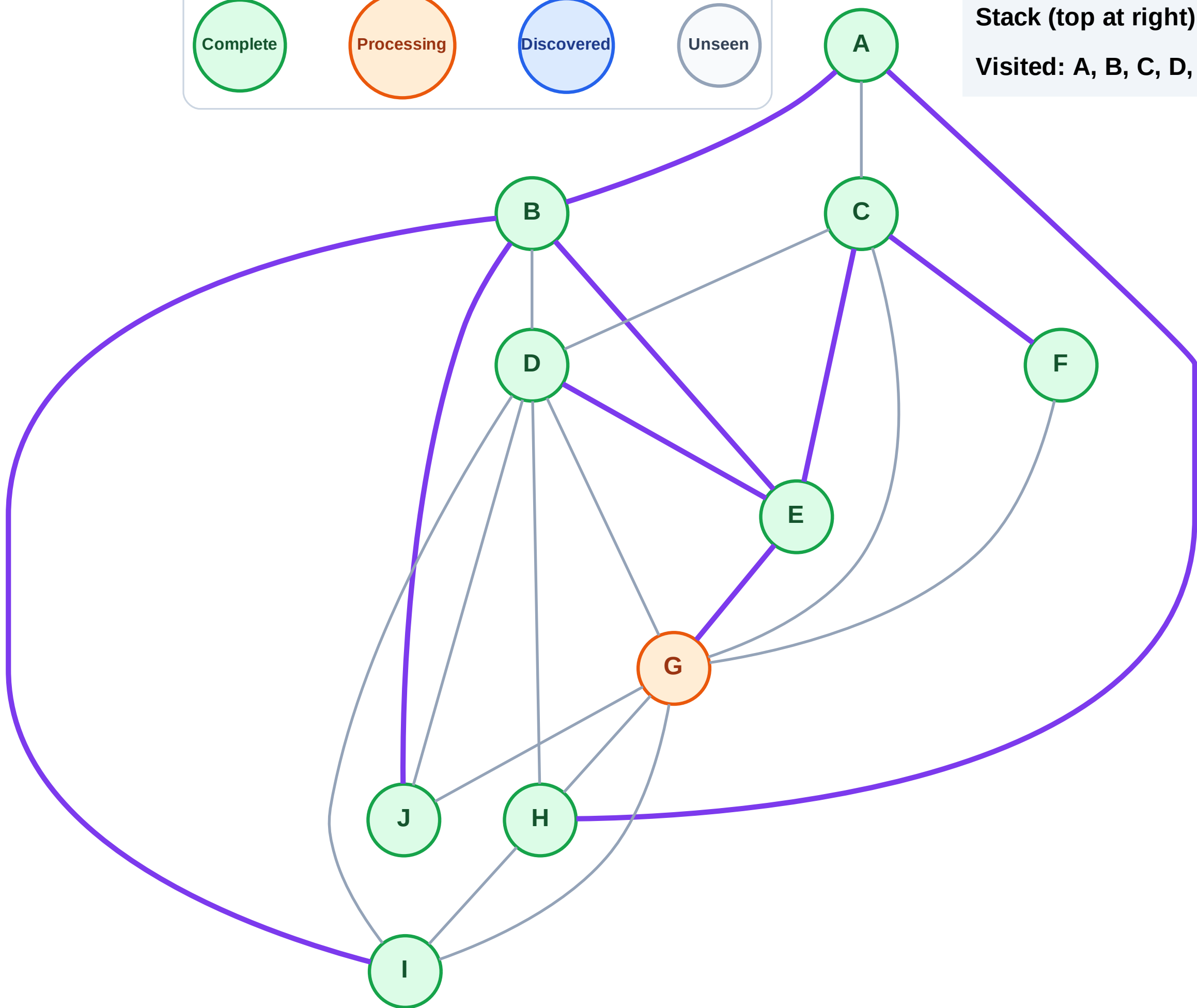
Step 20: Process node G

Legend



Stack (top at right): (empty)

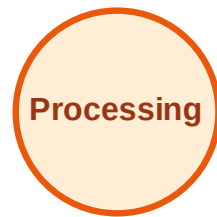
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

